

improve nuclease resistance, binding affinity, pharmacokinetics and biological activity (Pichon and Hollenstein, 2024). However, modified nucleotides are often poor substrates for natural polymerases, and suitable reversible protecting groups are harder to design for non-natural sugar or backbone structures.

## Enzyme cascades and ligase-based strategies

The future of enzymatic oligonucleotide synthesis is unlikely to rely on polymerases alone. Enzyme cascades and ligase-based strategies offer complementary routes, particularly for modified therapeutic oligonucleotides. Pichon and Hollenstein (2024) discuss an isothermal biocatalytic method in which a modified oligonucleotide is generated from a self-priming hairpin template using a polymerase and an endonuclease. This cascade can produce clinically relevant oligonucleotides at up to 100 mg scale and has a reported process mass intensity of 530 kg raw material per kg product, compared with more than 4000 kg for phosphoramidite synthesis (Pichon and Hollenstein, 2024).

Ligases also provide a promising alternative because they join shorter oligonucleotide fragments rather than adding one nucleotide at a time. DNA and RNA ligases can tolerate some nucleobase, sugar and backbone modifications, making them useful for assembling longer or heavily modified sequences (Pichon and Hollenstein, 2024). This is particularly relevant for oligonucleotide therapeutics, where the target molecule may contain many precisely positioned modifications. In the longer term, hybrid systems combining polymerases, ligases, endonucleases, microarrays and automated reagent delivery may offer the most flexible approach to manufacturing diverse oligonucleotide products.

The evolution of enzymatic oligonucleotide synthesis reflects a broader shift from purely chemical manufacturing toward controlled biocatalysis. Phosphoramidite chemistry remains indispensable because of its reliability, modification compatibility and industrial maturity. However, enzymatic synthesis offers attractive advantages, including milder aqueous conditions, potential reduction in hazardous waste, compatibility with automation and the ability to access longer sequences (Pichon and Hollenstein, 2024; Zhou et al., 2025).

At present, controlled enzymatic DNA synthesis is the most advanced area, especially through TdT-based systems using reversible terminators or polymerase–nucleotide conjugates. RNA synthesis is emerging rapidly but still faces challenges in protecting-group design and polymerase compatibility. XNA and heavily modified therapeutic oligonucleotides

remain the most difficult frontier. Future progress will likely depend on three converging advances: better reversible blocking groups, more highly engineered polymerases and integration with high-throughput synthesis technologies such as inkjet printing, chips, photochemistry and electrochemistry. As Zhou et al. (2025) suggest, combining enzymatic methods, microarrays and digital technologies may create the next generation of DNA synthesis platforms, while Pichon and Hollenstein (2024) argue that combining polymerase- and ligase-based strategies could ultimately enable the production of oligonucleotides of many types, lengths and modification patterns.

## Discovery

High-throughput screening is central to modern therapeutic discovery because it allows large numbers of molecular candidates to be tested rapidly for binding, activity, specificity, delivery, and safety. In RNA therapeutics, screening strategies vary widely depending on the modality: aptamers are enriched from vast randomized nucleic-acid libraries, oligonucleotide drugs are selected from sequence-designed panels, mRNA therapeutics require optimization of both construct and delivery system, and antibody discovery relies on large immune or display libraries. Although these platforms differ in chemistry and workflow, they share a common goal: converting enormous molecular diversity into a small set of candidates with the potency, selectivity, manufacturability, and biological performance needed for therapeutic development.

## Aptamers

SELEX is a high-throughput screening method for identifying DNA or RNA aptamers from very large randomized nucleic-acid libraries. In a typical SELEX screen, the starting library contains approximately  $10^{13}$ – $10^{15}$  unique oligonucleotide sequences, meaning that trillions of candidate binders can be screened in parallel against a target molecule (Li et al., 2023). The target may be a purified protein, small molecule, whole cell, virus, tissue sample, or immobilized ligand. Binding sequences are retained, while non-binding sequences are removed by washing, filtration, magnetic-bead capture, affinity chromatography, nitrocellulose partitioning, or capillary electrophoresis (Li et al., 2023).

After each binding round, retained sequences are amplified by PCR for DNA aptamers or by reverse transcription, PCR and transcription for RNA aptamers. Screening stringency is increased over successive rounds by reducing target concentration, shortening incubation time, increasing wash intensity, or adding competitor molecules. Counter-screening can also be used to remove aptamers that bind related but unwanted targets. Modern high-throughput SELEX often combines the enrichment process with next-generation sequencing, allowing researchers to track millions of sequence reads and identify enriched aptamer families rather than relying only on endpoint cloning (Darmostuk et al., 2015; Li et al., 2023).

For synthesis, SELEX libraries are generally produced as randomised DNA oligonucleotide pools by commercial oligo vendors. Examples include IDT oPools, which support pooled DNA oligos of 40–350 bases and up to 20,000 oligos per pool, and Twist Bioscience Oligo Pools, which use silicon-based DNA synthesis to generate hundreds of thousands of oligos per run for complex screening libraries (IDT, 2019; Twist Bioscience, 2026). Custom SELEX services are also offered by specialist CROs such as Creative Biolabs, CD Bioparticles and Novaptech, which provide aptamer screening against proteins, cells and other target formats.

### RNA-targeting therapeutic screening

High-throughput screening of RNA-targeting therapeutics depends on whether the candidates are oligonucleotide drugs or small molecules that bind RNA. For oligonucleotide drugs such as siRNAs, antisense oligonucleotides, splice-switching oligonucleotides and microRNA mimics or inhibitors, screening usually begins with computational design of large sequence panels complementary to the target RNA; La Jolla Labs PTx software is a common platform for design. These panels may range from dozens to hundreds of candidates for a single transcript to genome-wide RNAi libraries covering tens of thousands of genes.

These candidates are usually screened in multi-well plate formats after transfection into cells. The readouts include qPCR, RT-qPCR, RNA sequencing, reporter-gene activity, western blotting, ELISA, fluorescence imaging, high-content microscopy and cell-viability assays. Hits are selected based on knockdown efficiency, splice correction, protein reduction, potency, durability and low cytotoxicity. Secondary screens assess off-target effects, immune stimulation, transcriptome-wide specificity and activity in disease-relevant cell models (Roberts, Langer and Wood, 2020; Hofman and Corey, 2023). Commercial RNAi screening libraries and synthesis platforms are available from Thermo Fisher Scientific, Horizon Discovery/Revvity, MilliporeSigma MISSION, Dharmacon/Horizon, Qiagen and other

oligonucleotide vendors; these vendors offer predesigned, custom, arrayed or pooled siRNA, shRNA and microRNA libraries for high-throughput functional genomics screens (Thermo Fisher Scientific, 2026; Horizon Discovery, 2026; MilliporeSigma, 2026).

For small molecules that target RNA, the screen size depends heavily on the platform. Conventional biochemical or cell-based high-throughput screening often tests thousands to tens of thousands of compounds. For example, Life Chemicals markets RNA-focused screening libraries containing more than 29,600 drug-like compounds, while other HTS collections such as ChemBridge or Maybridge commonly contain around 50,000 compounds or more (Life Chemicals, 2025; Thermo Fisher Scientific, 2026; Northwestern HTAL, 2026). Alternative methods such as small-molecule microarrays can screen immobilised compound libraries against labelled RNA targets; Arrayjet's SMM-10 platform contains 10,000 pre-selected compounds (Arrayjet, 2026). At the ultra-large scale, DNA-encoded libraries can screen millions to billions of small molecules in pooled affinity-selection formats, and DEL approaches have been applied to RNA targets such as HIV-1 TAR RNA (Peterson and Liu, 2023; Chen et al., 2022).

### mRNA therapeutic screening

High-throughput screening for mRNA therapeutics usually focuses on two linked screening problems: screening the mRNA construct and screening the delivery formulation. Construct screening compares mRNA designs that vary in 5' cap structure, 5' and 3' untranslated regions, coding-sequence codon optimization, modified nucleosides, open reading frame design, poly(A) tail length and purification method. These variants are tested in cell-based assays to measure protein expression, duration of expression, mRNA stability, innate immune activation and cytotoxicity (Sahin, Karikó and Türeci, 2014; Hou et al., 2021).

The largest screening bottleneck is often the delivery system, especially lipid nanoparticles. Automated formulation platforms can screen hundreds to thousands of lipid nanoparticle formulations. One published high-throughput mRNA-LNP platform reports screening up to 384 formulations per plate and states that several plates per run can support comparison of hundreds or thousands of novel LNPs (Lokugamage et al., 2022). Another in vivo barcoded LNP study screened over 400 nanoparticle formulations containing 384 unique ionisable lipids in only nine mice, using peptide-encoding mRNA barcodes and LC-MS/MS readout (Rhym et al., 2023).

Common technologies include automated liquid handling, microfluidic LNP mixing, reporter mRNA expression assays, luciferase or fluorescent-protein readouts, flow cytometry, high-content imaging, cytokine assays, LC-MS/MS barcode quantification and in vivo biodistribution studies. Vendors and platforms used in this space include TriLink BioTechnologies and GENEWIZ/Azenta for custom mRNA synthesis, Cytiva/Precision NanoSystems NanoAssemblr systems for microfluidic nanoparticle formulation, PreciGenome NanoGenerator Flex-S Plus, which advertises 96 samples per hour, and Unchained Labs Sunscreen, which automates screening of up to 96 formulations (TriLink BioTechnologies, 2026; Azenta, 2026; PreciGenome, 2026; Unchained Labs, 2026).

## Antibody screening

High-throughput antibody screening is used to identify antibodies with the desired binding, specificity and biological function. Screening can begin with hybridomas, immunised-animal B cells, human B-cell repertoires, single-cell B-cell sequencing, phage display, yeast display, mammalian display or synthetic antibody libraries. In display-based antibody discovery, the number of antibody variants screened can be extremely large. Modern phage-display libraries commonly contain around  $10^{10}$ – $10^{11}$  antibody variants, with published therapeutic antibody libraries ranging from  $1.5 \times 10^{10}$  to  $3.6 \times 10^{11}$  variants (Almagro et al., 2019).

In phage display, antibody fragments such as scFv or Fab molecules are displayed on bacteriophage particles and panned against immobilized or solution-phase antigen. Non-binding phage are removed, while antigen-binding phage are recovered, amplified in bacteria and subjected to additional rounds of selection. Yeast display and mammalian display allow antibody binding to be screened by fluorescence-activated cell sorting, often enabling selection by antigen-binding strength, expression level and cross-reactivity. After primary screening, individual hits are tested by ELISA, flow cytometry, surface plasmon resonance, biolayer interferometry, homogeneous fluorescence assays or cell-based functional assays (Hoogenboom, 2005; Almagro et al., 2019).

Secondary antibody screens measure affinity, epitope binning, receptor blocking, ligand neutralisation, immune-cell recruitment, antibody-dependent cellular cytotoxicity, complement activation, internalisation, species cross-reactivity and developability. Developability screening often includes expression titre, aggregation risk, thermal stability, hydrophobicity, polyspecificity and manufacturability. Vendors and technologies commonly used include GenScript for high-throughput gene-to-antibody production,

Sino Biological and Creative Biolabs for phage-display and antibody-discovery services, Carterra LSA, Sartorius Octet BLI, Cytiva Biacore SPR, Berkeley Lights/Bruker Beacon and Sphere Fluidics Cyto-Mine for high-throughput antibody characterisation or single-cell screening. GenScript describes high-throughput gene-to-antibody production using high-throughput gene synthesis and transient antibody expression, with standard projects completed in as few as 18 business days (GenScript, 2026).

## Chemistry, Manufacturing, and Controls (CMC)

The chemistry, manufacturing, and controls (CMC) framework for RNA therapeutics must accommodate two related but distinct development paradigms. For large indications, the CMC program must ultimately support reproducible, scalable, commercial manufacturing, validated analytical methods, justified specifications, long-term stability, process robustness, and lifecycle management. For ultra-rare and N-of-1 therapies, the CMC program may be more compressed and risk-based, particularly where the ASO belongs to a well-characterized chemical class, is manufactured using conventional methods, and is formulated as a simple aqueous solution or lyophilized powder for reconstitution (FDA, 2021). In both settings, however, the central CMC objective is the same: to ensure that the ASO drug substance and drug product are consistently identifiable, pure, potent, stable, and suitable for clinical administration.

### Simple ASOs without conjugation

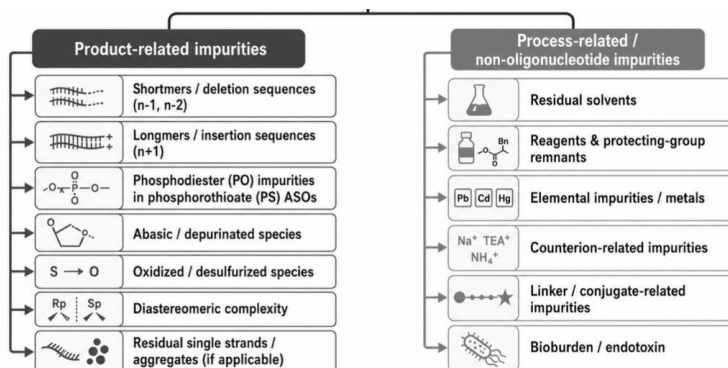
ASO CMC begins with precise definition of the drug substance. Unlike many small molecules, an ASO cannot be fully described by molecular formulas alone. The sequence, chain length, backbone chemistry, sugar modifications, nucleobase composition, terminal modifications and salt form must be defined and controlled. For phosphorothioate-containing ASOs, stereochemistry is also an important consideration because each phosphorothioate linkage introduces a chiral phosphorus center. Conventional phosphorothioate ASOs are often manufactured as reproducible mixtures of diastereomers rather than as single stereochemically pure entities. Consequently, the CMC control strategy must demonstrate consistency of the resulting product profile rather than necessarily resolving every stereoisomer (EMA, 2024).

Most ASOs are manufactured by solid-phase oligonucleotide synthesis, commonly using phosphoramidite chemistry. The standard synthetic cycle includes deprotection or detritylation, coupling of a protected nucleotide building block, oxidation or sulfurization to generate the intended internucleotide linkage, and capping of unreacted chains. Following chain assembly, the oligonucleotide is cleaved from the solid support and deprotected to generate the crude product. The core drug substance workflow then typically proceeds through purification, isolation, and, where applicable, lyophilization or preparation of the active substance in solution (Chen et al., 2026; EMA, 2024). Common purification approaches include anion-exchange chromatography, ion-pair reversed-phase chromatography, hydrophobic interaction chromatography,

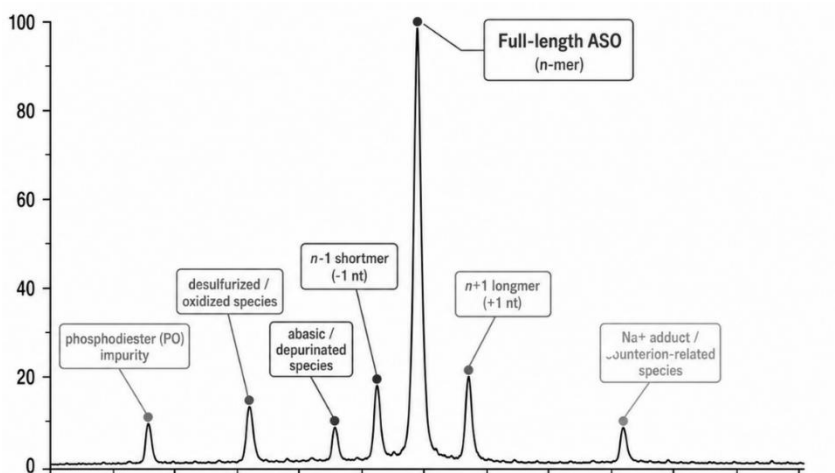
ultrafiltration/diafiltration, desalting, and combinations of orthogonal methods.

A major feature of ASO CMC is the need to understand and control both product-related and process-related impurities. Product-related impurities may include shortmers, longmers, deletion sequences, insertion sequences, phosphodiester impurities in phosphorothioate ASOs, depurinated or abasic species, oxidized or desulfurized products, cross-linked species, and degradation products formed during manufacture or storage. Non-oligonucleotide impurities may include residual solvents, reagents, protecting groups, elemental impurities, counterion-related impurities, and impurities derived from linkers, solid supports, or conjugation chemistry. Because many ASO-related impurities are structurally similar to the parent molecule, and because some may co-elute or differ only subtly in mass or sequence position, a single analytical method is rarely sufficient to support a complete control strategy (EMA, 2024).

Analytical characterization is central to ASO development. Mass spectrometry, including LC-MS and tandem MS, is commonly used for molecular weight determination, sequence confirmation, and impurity identification. Chromatographic methods such as ion-pair reversed-phase HPLC, anion-exchange HPLC, and other orthogonal separation methods are used for purity, related substances, assay, and in-process controls. Additional techniques may include UV spectroscopy, melting temperature analysis, NMR, capillary electrophoresis, elemental analysis, Karl Fischer water determination, GC for residual solvents, and compendial microbiological and endotoxin methods. For drug product, especially injectable presentations, CMC controls extend to pH, osmolality, visible and subvisible particulates, sterility assurance, container-closure integrity, extractables and leachables, and compatibility with the final dosage form (FDA, 2021; EMA, 2024).



For large-population ASO therapeutics, CMC development typically matures progressively from early clinical supply to a robust commercial control strategy. Early-stage development may rely on platform knowledge, qualified methods, provisional specifications, and limited stability data sufficient to support clinical investigation. As development advances toward registration, the process must become more product-specific and better characterized. This includes identification of critical quality attributes, understanding of critical process parameters, establishment of in-process controls, validation of analytical procedures, justification of specifications, generation of representative batch data, stability studies supporting shelf life, and demonstration that the intended commercial manufacturing process can consistently deliver material of appropriate quality (Chen et al., 2026; EMA, 2024).



For ultra-rare and N-of-1 ASO therapies, the CMC paradigm is more flexible but still quality-driven. FDA’s draft guidance for individualized ASO drug products focuses on therapies for severely debilitating or life-threatening diseases caused by unique genetic variants where only one or two individuals are prospectively identified. Within this context, the guidance emphasizes well-characterized ASO chemical classes, conventional manufacturing methods, and relatively simple formulations, while recommending CMC information sufficient to support first-in-human exposure (FDA, 2021). Such information includes the description of the drug substance, manufacturing process, characterization, drug substance and drug product controls, reference standards or materials, container closure systems, and stability. Thus, even when development timelines are

compressed, the CMC package must still provide confidence in identity, quality, purity, strength, and patient safety.

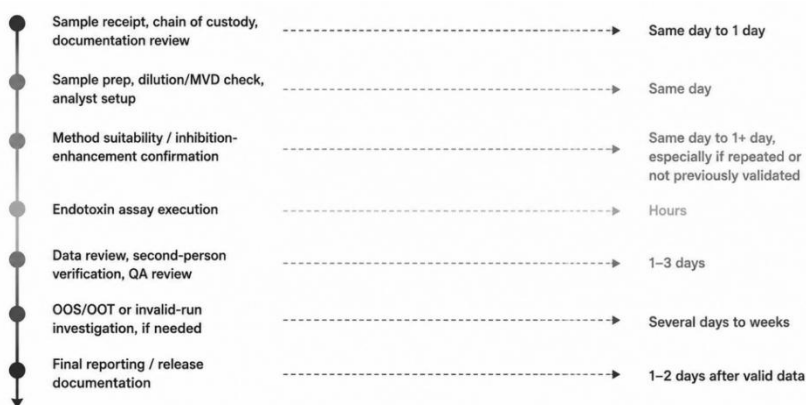
The practical difference between commercial ASO CMC and individualized ASO CMC is therefore primarily one of scale, timing, and evidentiary depth. A large-indication ASO program must support reproducible manufacturing across many batches, potentially multiple sites, and a long commercial lifecycle. An individualized ASO program must prioritize speed, use of prior knowledge, focused risk assessment, and sufficient analytical confidence to justify administration to one or a very small number of patients. In both cases, ASO CMC requires integration of sequence-defined molecular design, controlled chemical synthesis, impurity understanding, orthogonal analytical methods, appropriate specifications, formulation and container-closure controls, and stability assessment. The goal is to ensure that every ASO, whether developed for a broad population or a single patient, is manufactured and controlled to a standard appropriate for its clinical context.

Bioburden control for antisense oligonucleotides (ASOs) manufactured under GMP is an important element of the overall contamination control strategy. Because ASOs may be produced through multi-step synthesis, purification, concentration, and formulation processes, microbial control should be maintained through qualified raw materials, controlled water systems, validated cleaning procedures, environmental monitoring, closed or appropriately protected processing, and defined in-process hold conditions. Bioburden testing should be performed at appropriate process points based on risk, particularly before sterile filtration or final aseptic processing, to confirm that microbial levels remain within established alert, action, and acceptance limits. Results should be trended and investigated when excursions occur, with corrective actions documented under the quality system. For sterile ASO drug products, bioburden control supports—but does not replace—validated sterilization, aseptic processing controls, endotoxin control, and final sterility assurance.

## Endotoxin

Endotoxin control is a critical quality attribute for ASOs intended for parenteral or intrathecal administration. Endotoxin levels should be controlled through qualified raw materials, controlled water systems, appropriate equipment cleaning, validated depyrogenation or endotoxin-reduction steps where applicable, and routine in-process and release testing. Testing should be performed using a compendial bacterial endotoxins test method, with method suitability demonstrated for the ASO matrix to address

potential assay interference. Endotoxin acceptance criteria should be justified based on the intended route of administration, maximum dose, patient population, and applicable regulatory or pharmacopeial expectations.



Endotoxin testing should be performed using a validated bacterial endotoxins test method suitable for the ASO drug substance, intermediate, or drug product being evaluated. The assay should be conducted using qualified reagents, calibrated equipment, endotoxin-free consumables, and controlled test conditions to prevent false-positive or false-negative results. Samples should be handled aseptically and diluted, where necessary, using endotoxin-free water to bring the test article within the validated assay range and to reduce any product-related interference. The selected method may be gel-clot, chromogenic, or turbidimetric, provided that method suitability has been demonstrated for the specific ASO matrix.

Prior to routine sample testing, method suitability should be confirmed by demonstrating that the ASO sample does not inhibit or enhance endotoxin detection at the proposed dilution. This is typically performed by testing unspiked sample preparations alongside sample preparations spiked with a known amount of control standard endotoxin. Recovery of the endotoxin spike should fall within the predefined acceptance range specified by the applicable pharmacopeial method and site procedure. The maximum valid dilution should be calculated based on the endotoxin limit, product concentration, and assay sensitivity, and routine testing should be performed at a dilution that is justified, validated, and documented.

For each assay run, appropriate controls should be included, such as reagent negative controls, positive product controls, endotoxin standard controls, and, where applicable, standard curve controls. The test is considered valid only if all system suitability and control criteria are met.

Sample results should be reported in endotoxin units per millilitre, endotoxin units per milligram, or another justified unit appropriate to the material and specification. Results exceeding the established acceptance criterion should be investigated under the quality system, including assessment of sample handling, reagent performance, assay interference, laboratory contamination, and potential process-related sources of endotoxin. Confirmatory or repeat testing should be scientifically justified and documented according to approved GMP procedures.

## Stability

Stability studies should be designed to support the proposed storage conditions, shelf life, and in-use handling of the ASO drug substance and drug product. GMP stability programs typically include long-term, accelerated, and, where appropriate, stress or forced-degradation studies to evaluate chemical, physical, and microbiological stability. Stability-indicating analytical methods should monitor relevant ASO quality attributes, such as identity, purity, assay, degradation products, water content, appearance, pH, concentration, particulate matter, sterility or microbial quality, and endotoxin as applicable. Data should be trended over time, and any out-of-specification or significant changes should be investigated under the quality system. The stability program should also assess container-closure compatibility, freeze-thaw sensitivity, light exposure, and hold times for intermediates, drug substance, and formulated drug product where relevant.

Antisense oligonucleotide (ASO) stability testing should evaluate both the intrinsic chemical stability of the molecule and its susceptibility to enzymatic degradation. ASOs are short, single-stranded synthetic nucleic acid therapeutics that rely on sequence-specific binding to RNA targets; however, unmodified oligonucleotides are vulnerable to degradation by endonucleases and exonucleases. As a result, most therapeutic ASOs incorporate chemical modifications, such as phosphorothioate backbones and modified sugar chemistries, to improve nuclease resistance, binding affinity and pharmacological performance (Crooke et al., 2018; Collotta et al., 2023). Stability testing should therefore include stability-indicating analytical methods capable of detecting degradation products, truncations, depurination, desulfurisation, oxidation, aggregation or other relevant impurities, as well as changes in assay, purity, appearance, pH, water content and concentration where applicable.

For GMP development, ASO stability studies should support the proposed storage conditions, retest period or shelf life for both drug

substance and drug product. Programs commonly include long-term and accelerated studies, together with forced-degradation or stress studies to demonstrate method specificity and identify likely degradation pathways. Because the primary degradation pathway for ASOs is often nuclease-mediated cleavage, in vitro stability studies using biological matrices or defined nuclease systems can provide useful supporting information on how sequence and chemical modifications influence metabolic stability (Sutton et al., 2025). For formulated ASO drug products, stability protocols should also consider container-closure compatibility, adsorption or leachables risk, freeze-thaw exposure, light sensitivity, in-use hold times, particulate matter, sterility or bioburden, and endotoxin where relevant to the route of administration.

|                                 |                                                                                               |                            |                                                                                                                                           |
|---------------------------------|-----------------------------------------------------------------------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Document No.:</b>            | SPEC-DP-ASO1-001                                                                              | <b>Effective Date:</b>     | TBD                                                                                                                                       |
| <b>Version:</b>                 | 0.1 Draft                                                                                     | <b>Specification Type:</b> | GMP release and stability specification                                                                                                   |
| <b>Product Name:</b>            | ASO1 Sterile Injection                                                                        | <b>Status:</b>             | Draft template – Not approved for GMP release until product-specific validation, stability, toxicology and regulatory review are complete |
| <b>Drug Substance:</b>          | ASO1 sodium salt, single-stranded antisense oligonucleotide                                   |                            |                                                                                                                                           |
| <b>Dosage Form:</b>             | Sterile aqueous solution for injection                                                        |                            |                                                                                                                                           |
| <b>Strength:</b>                | TBD mg/mL ASO1                                                                                |                            |                                                                                                                                           |
| <b>Fill Volume:</b>             | TBD mL per vial                                                                               | <b>Container Closure:</b>  | USP Type I glass vial with sterile elastomeric stopper and aluminum seal, or equivalent                                                   |
| <b>Route of Administration:</b> | TBD (e.g., intrathecal, subcutaneous, intravitreal)                                           |                            |                                                                                                                                           |
| <b>Storage Condition:</b>       | Store frozen or refrigerated as justified by stability data; protect from light if applicable |                            |                                                                                                                                           |

| No. | TEST / QUALITY ATTRIBUTE            | METHOD                              | ACCEPTANCE CRITERION                                                                                                                             | RELEASE     | STABILITY   |
|-----|-------------------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
| 1   | Appearance                          | Visual inspection                   | Clear, colorless to slightly opalescent solution; essentially free from visible particulates; no discoloration                                   | Yes         | Yes         |
| 2   | Identity: Molecular mass            | ESI-MS (LC-MS)                      | Observed MW conforms to 7154.5 Da $\pm$ 2.0 Da                                                                                                   | Yes         | As needed   |
| 3   | Identity: Chromatographic retention | IE-HPLC or IP-RP-HPLC               | Retention time of principal peak conforms to reference standard                                                                                  | Yes         | Yes         |
| 4   | Identity: Sequence confirmation     | LC-MS/MS or other orthogonal method | Sequence confirmed or bridged to reference standard / drug substance lot                                                                         | Yes         | As needed   |
| 5   | Assay / Strength                    | UV at 260 nm or HPLC-UV             | 90.0 – 110.0% of label claim                                                                                                                     | Yes         | Yes         |
| 6   | Full-length ASO content             | Stability-indicating HPLC/UPLC      | Report full-length content: NLT 90.0% (or product-specific)                                                                                      | Yes         | Yes         |
| 7   | Overall purity                      | IE-HPLC or IP-RP-HPLC               | NLT 90.0% total ASO purity                                                                                                                       | Yes         | Yes         |
| 8   | Individual specified impurities     | Stability-indicating HPLC/UPLC      | Each specified impurity: NMT 1.5% (or product-specific)                                                                                          | Yes         | Yes         |
| 9   | Unspecified individual impurities   | Stability-indicating HPLC/UPLC      | Each unspecified impurity: NMT 1.0%                                                                                                              | Yes         | Yes         |
| 10  | Total impurities / degradants       | Stability-indicating HPLC/UPLC      | NMT 10.0% (or product-specific)                                                                                                                  | Yes         | Yes         |
| 11  | Phosphodiester impurity             | LC-MS or HPLC                       | NMT 1.0% (or product-specific)                                                                                                                   | Yes         | Yes         |
| 12  | Aggregates / higher order species   | SEC-HPLC or non-denaturing HPLC     | Report: NMT 2.0% (or product-specific)                                                                                                           | Conditional | Conditional |
| 13  | pH                                  | USP <791>                           | 6.5 – 8.0 (or product-specific)                                                                                                                  | Yes         | Yes         |
| 14  | Osmolality                          | USP <785>                           | 250 – 400 mOsm/kg                                                                                                                                | Yes         | Yes         |
| 15  | Particulate matter, subvisible      | USP <788>                           | Meets USP <788> for small-volume injections                                                                                                      | Yes         | Yes         |
| 16  | Visible particulate matter          | USP <790>                           | Meets compendial requirements                                                                                                                    | Yes         | Yes         |
| 17  | Sterility                           | USP <71>                            | No growth                                                                                                                                        | Yes         | Yes*        |
| 18  | Bacterial endotoxins                | USP <85>                            | NMT route- and dose-based limit:<br>• Intrathecal: NMT 0.2 EU/kg/hour<br>• Subcutaneous: NMT 5 EU/kg/hour<br>• Intravitreal: NMT 2.0 EU/dose/eye | Yes         | Yes         |

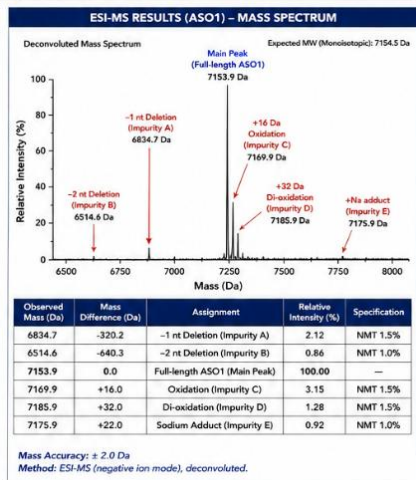
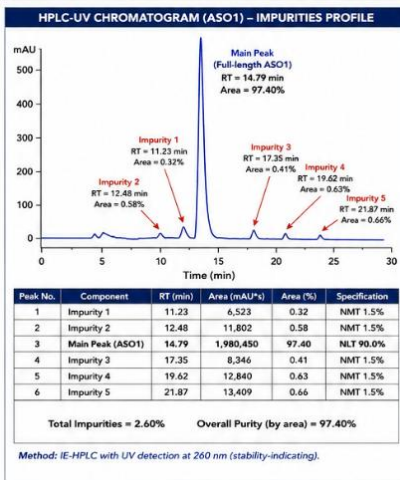
\*Sterility is a release test. For stability, sterility or container closure integrity (CCI) may be used as stability endpoints.

**ADDITIONAL NOTES:**

- Specifications are based on current knowledge and may be tightened as more data become available.
- Methods are stability-indicating and qualified/validated per ICH Q2(R2).
- Endotoxin limits are established based on maximum proposed dose and route of administration.
- This is a sample template and must be product-specifically justified and approved.

**BATCH RELEASE DECISION:**

A batch may be released only if all release tests meet acceptance criteria, all deviations are resolved, sterility assurance and CCI are met (if applicable), stability data support the proposed use period, and QA has approved the batch record and CoA.



## CMC for Lipid Nanoparticle Drug Products

Lipid nanoparticles (LNPs) are an increasingly important non-viral delivery platform for nucleic acid therapeutics, including mRNA, siRNA, gene-editing components, and potentially other oligonucleotide-based drug substances. Their role in drug development is not simply as passive excipients, but as functional delivery systems that protect nucleic acid cargo, support cellular uptake, influence biodistribution, and affect pharmacology,

safety, and stability. As a result, the chemistry, manufacturing, and controls (CMC) strategy for an LNP product must address both the nucleic acid cargo and the nanoparticle delivery system as an integrated drug product (Cameau et al., 2022; Johnson, Ho and Kapatrala, 2024).

A typical LNP CMC process begins with definition and control of raw materials. These include the nucleic acid drug substance, ionizable lipid, helper phospholipid, cholesterol, PEG-lipid, ethanol or other organic solvents, aqueous buffers, water for injection, filters, tubing, single-use assemblies, and container-closure components. Lipid excipients require particular attention because their purity, impurity profile, safety, and consistency can significantly affect the quality of the finished nanoparticle. Novel or non-compendial lipids may require substantial CMC and safety justification, including impurity evaluation, residual solvent control, elemental impurity risk assessment, genotoxic impurity assessment, and documentation of manufacturing controls. (MilliporeSigma, 2026).

LNP formation is commonly achieved by rapid mixing of a lipid solution in ethanol with an aqueous nucleic acid solution under controlled conditions. During this process, the lipids self-assemble around the nucleic acid cargo to form nanoparticles. Critical process parameters may include lipid concentration, nucleic acid concentration, lipid-to-cargo ratio, N/P ratio, aqueous-to-organic flow ratio, total flow rate, mixing geometry, temperature, buffer composition, pH, ionic strength, and hold times. Because particle size, polydispersity, morphology, surface charge, encapsulation efficiency, and potency are sensitive to the mixing environment, LNP manufacturing requires tight process control and scalable equipment that can maintain comparable mixing conditions from small-scale formulation screening through clinical and commercial production (Cameau et al., 2022).

After nanoparticle formation, the process typically proceeds through dilution, concentration, diafiltration, solvent removal, buffer exchange, sterile filtration, and fill-finish. Tangential flow filtration is commonly used to remove ethanol and unencapsulated small molecules while concentrating the LNPs into the final formulation buffer. These downstream steps can themselves influence product quality because LNPs may be sensitive to shear, temperature, filter type, membrane chemistry, processing time, and freeze-thaw exposure. Sterile filtration and filling must also be carefully evaluated because nanoparticle products can be affected by filter compatibility, adsorption, particle-size shifts, visible or subvisible particles, and container-closure interactions. For some products, frozen storage, refrigerated storage, or lyophilization may be considered, but each approach requires product-specific stability justification (Cameau et al., 2022; Jeffs, Ho and Kapatrala, 2024).

The analytical control strategy for LNPs is necessarily multidimensional. In addition to conventional drug product tests such as appearance, identity, assay, pH, osmolality, sterility, endotoxin, subvisible particles, residual solvents, and container-closure integrity, LNP-specific characterization is required. Common critical quality attributes include particle size, polydispersity index, encapsulation efficiency, nucleic acid integrity, lipid identity and content, lipid impurity profile, residual ethanol, surface charge or zeta potential, potency, in vitro expression or functional activity, and stability-indicating degradation products. Because standardized nanomaterial methods are still evolving, complementary analytical methods and carefully justified sample preparation, sampling, and method-suitability strategies are especially important (Stephens, 2022; Cameau et al., 2022).

CMC development for LNPs must also address manufacturing scale, automation, facility design, and data integrity. Scale-up from milliliter-scale formulation development to clinical or commercial-scale manufacturing can introduce variability in mixing, temperature control, residence time, solvent handling, TFF performance, and sterile filtration. For this reason, scale-down models, automated platform technologies, single-use systems, electronic batch records, real-time monitoring, and robust process characterization are frequently emphasized as tools for reducing variability and improving GMP readiness. (Johnson, Ho and Kapatrala, 2024; Jeffs, Ho and Kapatrala, 2024).

The nanomaterial nature of the product does not create an entirely separate quality standard, but it does increase the need to characterize how the nanoparticle's physical and chemical attributes relate to safety, efficacy, and quality. Developers are expected to identify critical quality attributes, understand process risks, justify analytical methods, bridge formulation or process changes, retain samples for comparability, and maintain lifecycle control. Early interaction with regulators is particularly important because the regulatory framework for LNPs continues to evolve, and each formulation may have unique features depending on lipid composition, cargo type, route of administration, dose, therapeutic area, and intended patient population (Stephens, 2022; Johnson, Ho and Kapatrala, 2024).

Compared with naked ASO drug products, LNP-based products introduce a major additional layer of CMC complexity. A naked ASO formulation may often be developed as a relatively simple aqueous or lyophilized oligonucleotide product, with the primary CMC burden focused on ASO identity, purity, impurity profile, assay, microbiological quality, endotoxin, stability, and container compatibility. By contrast, an LNP product must control not only the nucleic acid cargo, but also multiple lipid components, lipid impurities, nanoparticle assembly, particle size distribution, encapsulation efficiency, residual solvent, surface properties,

filterability, physical stability, and interactions among cargo, lipids, process materials, and container closure. This greatly increases analytical burden, process-development work, raw-material qualification, facility requirements, regulatory justification, and comparability testing. In practical terms, LNPs can improve delivery and enable therapeutic applications that naked ASOs may not achieve, but they usually add substantial cost, time, technical risk, and operational complexity to CMC development and GMP manufacturing.

## Conclusion

RNA therapeutics began with a simple but transformative idea: that nucleic acids are not only carriers of biological information, but programmable molecules that can be designed, synthesized, modified, delivered, and translated into clinical action.

The history of this field is one of iteration. Each generation of RNA therapeutic technology solved one problem and exposed the next. Early antisense oligonucleotides showed that sequence-specific recognition could alter gene expression, but they also revealed the limitations of unmodified nucleic acids: instability, rapid clearance, poor tissue access, and nonspecific toxicity. Phosphorothioate backbones improved nuclease resistance and pharmacokinetics, but introduced new questions around protein binding, inflammation, and tolerability. Sugar modifications such as 2'-O-methyl, 2'-fluoro, 2'-MOE, locked nucleic acids, constrained ethyl chemistries, and newer analogues expanded the medicinal chemistry toolkit, allowing potency, stability, affinity, and safety to be tuned with increasing precision. Splice-switching oligonucleotides showed that RNA could be remodeled rather than simply destroyed. siRNA therapeutics harnessed endogenous RNA interference to make gene silencing catalytic and durable. mRNA therapeutics reversed the logic entirely, using synthetic RNA not to suppress a message, but to introduce one.

These advances did not arise from a single breakthrough. They emerged through disciplined cycles of design, synthesis, testing, failure, redesign, and translation. The first molecules were crude by today's standards, but they established the central principle that still defines the field: RNA therapeutics are information medicines made possible by chemistry. The sequence provides specificity, but chemistry determines whether that sequence can survive, distribute, enter cells, engage the correct machinery, avoid the wrong biology, and produce a therapeutic effect in patients.

This iterative progress is enabling a rational form of drug design. In traditional small-molecule discovery, screening often begins with large chemical libraries and uncertain target engagement. In RNA therapeutics, the starting point can be genetic information itself: a pathogenic mutation, a toxic transcript, a splice defect, a disease-associated isoform, a regulatory RNA, or a protein-coding message that needs to be supplied. Once the biological hypothesis is defined, therapeutic candidates can be designed

directly against the relevant sequence or regulatory element. Screening is still essential, but it becomes more targeted. Instead of searching broadly for activity, the field can focus on optimizing chemistry, position, modification pattern, delivery format, potency, selectivity, durability, and safety.

This changes the economics and tempo of discovery. RNA therapeutics can compress the distance between genetic diagnosis and therapeutic hypothesis. Candidate panels can be designed in silico, synthesized rapidly, and screened in disease-relevant systems, including patient-derived cells. Iterative chemistry allows weak leads to be improved systematically rather than abandoned blindly. Off-target analysis, transcriptome profiling, allele selectivity, splice correction, pharmacodynamic biomarkers, and dose-response behavior can be incorporated early. The result is not that RNA drug discovery becomes easy. It does not. Rather, it becomes more rational, more modular, and more directly connected to the biology of disease.

The same logic is reshaping the path to the clinic. RNA therapeutics are increasingly developed through platform knowledge: shared chemistries, shared manufacturing processes, shared analytical methods, shared toxicology principles, and shared delivery strategies. A GalNAc-conjugated siRNA, a 2'-MOE gapmer ASO, a splice-switching oligonucleotide, or an LNP-formulated mRNA each carries lessons from previous molecules in the same class. Those lessons inform dose selection, tissue targeting, safety monitoring, pharmacodynamic endpoints, and chemistry, manufacturing, and controls. Each approved drug makes the next one less mysterious.

Yet the field's greatest lesson is also its warning: programmability is not enough. A sequence is not a drug. A potent molecule in a dish is not a medicine. A beautiful mechanism is not a clinical benefit. RNA therapeutics must pass through every practical gate of drug development: delivery, biodistribution, endosomal escape, intracellular trafficking, durability, tolerability, analytical characterization, scale-up, purity, stability, regulatory review, and patient access. The failures in the field are as instructive as the successes. They show that potency without delivery is insufficient, affinity without specificity can be dangerous, and target knockdown without clinical benefit is not enough.

This is why the next era of RNA therapeutics will belong not only to those who can design sequences, but to those who can engineer systems. Better conjugates will expand tissue reach beyond the liver. Improved lipid nanoparticles and alternative delivery vehicles will make mRNA and RNA editing more precise. New backbone and sugar chemistries will tune stability, protein binding, immune recognition, and intracellular behavior. Computational design will improve target selection and reduce off-target risk. Patient-derived models will make screening more disease-relevant.

Manufacturing innovation will determine whether personalized and nano-rare therapies can move from heroic exceptions to repeatable clinical practice.

The final measure of the field will not be its elegance. It will be execution. We can design medicines with unprecedented precision, but can we deliver them reliably, manufacture them reproducibly and bring them to patients quickly enough to matter?

The logistics are the hard part: chemistry, delivery, toxicology, manufacturing, clinical operations, regulatory alignment, and access. The future of RNA medicine will be built by those who drive this.

“Amateurs talk strategy. Professionals talk logistics.”

General Omar Bradley

## References

- Anderson, K.P. et al. (1996) Inhibition of human cytomegalovirus immediate-early gene expression by an antisense oligonucleotide. *Antimicrobial Agents and Chemotherapy*, 40, pp. 2004–2011.
- Azad, R.F. et al. (1993) Antiviral activity of a phosphorothioate oligonucleotide complementary to human cytomegalovirus RNA. *Antimicrobial Agents and Chemotherapy*, 37, pp. 1945–1954.
- Beaucage, S.L. and Caruthers, M.H. (1981) ‘Deoxynucleoside phosphoramidites: A new class of key intermediates for deoxypolynucleotide synthesis’, *Tetrahedron Letters*, 22(20), pp. 1859–1862.
- Bennett CF, Baker BF, Pham N, Swayze E, Geary RS. Pharmacology of antisense drugs. *Annu Rev Pharmacol Toxicol*. 2017;57:81–105.
- Bennett CF, Swayze EE. RNA targeting therapeutics: molecular mechanisms of antisense oligonucleotides. *Annu Rev Pharmacol Toxicol*. 2010;50:259–293.
- Braasch, D. A., and Corey, D. R. (2001). Locked nucleic acids (LNA): Fine-tuning the recognition of DNA and RNA. *Chemistry & Biology*, 8(1), 1–7.
- Crooke, S.T. (2017) ‘Molecular mechanisms of antisense oligonucleotides’, *Nucleic Acid Therapeutics*, 27(2), pp. 70–77.
- Crooke, S. T., et al. (2018). Antisense drug discovery and development technology considered in a pharmacological context. *Biochemical Pharmacology*, 153, 17–28.
- Crooke, S. T., Witztum, J. L., Bennett, C. F., & Baker, B. F. (2017). Antisense drug discovery and development technology. *Nat Rev Drug Discov.*, 16(6), 401–423.
- Crooke, S. T., Witztum, J. L., Bennett, C. F., & Baker, B. F. (2018). RNA-targeted therapeutics. *Cell Metab.*, 27(4), 714–739.
- Dimitrova, D. G. et al. (2019). RNA 2'-O-methylation (Nm) modification in human diseases. *Genes*, 10, 117.
- Eckstein F, Gindl H. Polyribonucleotides containing a phosphorothioate backbone. *Eur J Biochem*. 1970;13:558–564.
- Eckstein F. Phosphorothioates, essential components of therapeutic oligonucleotides. *Nucleic Acid Ther*. 2014;24(6):374–387.
- Egli, M. and Manoharan, M., 2023. Chemistry, structure and function of approved oligonucleotide therapeutics. *Nucleic Acids Research*, 51(6), pp.2529–2573.
- Graham, G.G. and Scott, K.F. (2005) Mechanism of action of paracetamol. *American Journal of Therapeutics*, 12(1), pp. 46–55.
- Havens, M.A. and Hastings, M.L. (2016) ‘Splice-switching antisense oligonucleotides as therapeutic drugs’, *Nucleic Acids Research*, 44(14), pp. 6549–6563.
- Hill, A. C., & Hall, J. (2023). The MOE modification of RNA: Origins and widescale impact on the oligonucleotide therapeutics field. *Liberation Serif Chimica Acta*, 106, e202200169.
- IDT (2023). 2'-O-Methyl RNA modification overview.
- Iwamoto N, Butler DCD, Svrzikapa N, Mohapatra S, Zlatev I, Sah DWY, et al. Control of phosphorothioate stereochemistry substantially increases the efficacy of antisense oligonucleotides. *Nat Biotechnol*. 2017;35(9):845–851.
- Khorana, H.G. (2024) Har Gobind Khorana. Biographical Memoir by P.A. Sharp and U.L. RajBhandary. Washington, DC: National Academy of Sciences.

- Koshkin, A. A., Singh, S. K., Nielsen, P., Rajwanshi, V. K., Kumar, R., Meldgaard, M., Olsen, C. E., and Wengel, J. (1998). LNA (locked nucleic acids): Synthesis of the adenine, cytosine, guanine, 5-methylcytosine, thymine and uracil bicyclonucleoside monomers, oligomerisation, and unprecedented nucleic acid recognition. *Tetrahedron*, 54(14), 3607–3630.
- Kurreck, J. (2003) ‘Antisense technologies. Improvement through novel chemical modifications’, *European Journal of Biochemistry*, 270(8), pp. 1628–1644.
- Kurreck, J., Wyszko, E., Gillen, C., and Erdmann, V. A. (2002). Design of antisense oligonucleotides stabilized by locked nucleic acids. *Nucleic Acids Res.*, 30(9), 1911–1918.
- Lennox, K.A. and Behlke, M.A. (2016) ‘Chemical modification and design of antisense oligonucleotides’, *Gene Therapy*, 23(6), pp. 428–440.
- Lima WF, Rose JB, Nichols JG, Wu H, Migawa MT, Wyrzykiewicz TK, et al. Human RNase H1 discriminates between subtle variations in the structure of the heteroduplex substrate. *Mol Pharmacol*. 2007;71(1):83–91.
- Lin, T.-Y., Mehta, R., & Glatt, S. (2021). Pseudouridines in RNAs: switching atoms means shifting paradigms. *FEBS Letters*, 595, 2310–2322.
- Lovrić J, Yan J, Li XQ, Karlsborn T, Bood M, Dahlén A, et al. In vitro structure–activity relationship stability study of antisense oligonucleotide therapeutics using biological matrices and nucleases. *Pharmacol Res Perspect*. 2025;13:e70096.
- Ma, Y., Zhang, Z., Jia, B. and Yuan, Y. (2024) ‘Automated high-throughput DNA synthesis and assembly’, *Heliyon*, 10, e26967.
- Martin, P. (1995). A new access to 2'-O-alkylated ribonucleosides and properties of 2'-O-alkylated oligoribonucleotides. *Liberation Serif Chimica Acta*, 78(2), 486–504.
- Matsukura M, Zon G, Shinozuka K, et al. Inhibition of HIV replication by antisense oligodeoxynucleotides. *Proc Natl Acad Sci U S A*. 1987;84:7706–7710.
- Naveed, A., Cooper, J.A., Li, R., Hubbard, A., Chen, J., Liu, T., Wilton, S.D., Fletcher, S. and Fox, A.H. (2021) ‘NEAT1 polyA-modulating antisense oligonucleotides reveal opposing functions for both long non-coding RNA isoforms in neuroblastoma’, *Cellular and Molecular Life Sciences*, 78, pp. 2213–2230. Available at: <https://doi.org/10.1007/s00018-020-03632-6>
- Obika, S., Nanbu, D., Hari, Y., Morio, K., In, Y., Ishida, T., and Imanishi, T. (1998). Synthesis of 2'-O,4'-C-methyleneuridine and -cytidine. Novel bicyclic nucleosides having a fixed C3'-endo sugar puckering. *Tetrahedron Letters*, 39(30), 5401–5404.
- Pallan, P. S., et al. (2012). Structural insights into the conformational properties of constrained ethyl (cEt)-modified nucleic acids. *Nucleic Acids Res.*, 40(7), 3372–3382.
- Pallan, P. S., Allerson, C. R., Berdeja, A., Seth, P. P., Swayze, E. E., Prakash, T. P., and Egli, M. (2012). Structure and nuclease resistance of 2',4'-constrained 2'-O-methoxyethyl (cMOE) and 2'-O-ethyl (cEt) modified DNAs. *Chemical Communications*, 48(66), 8195–8197.
- Prakash, T.P. et al. (2008) ‘Comparing in vitro and in vivo activity of 2'-O-[2-(methylamino)-2-oxoethyl]- and 2'-O-methoxyethyl-modified antisense oligonucleotides’, *Journal of Medicinal Chemistry*, 51(9), pp. 2766–2776.
- Roberts, T.C. et al. (2024) ‘uORF-targeting steric block antisense oligonucleotides do not induce protein upregulation’, *Molecular Therapy – Nucleic Acids*.
- Sang, A., Zhuo, S., Bochanis, A., Manautou, J. E., Bahal, R., Zhong, X.-b., & Rasmussen, T. P. (2024). Mechanisms of action of the US Food and Drug Administration-approved antisense oligonucleotide drugs. *BioDrugs*, 38, 511–526.

- Seth, P. P., et al. (2010). Second-generation antisense oligonucleotides: design, mechanism, and clinical application. *Current Opinion in Molecular Therapeutics*, 12(3), 289–298.
- Seth PP, Swayze EE. The medicinal chemistry of RNase H-activating antisense oligonucleotides. In: Wan Y, editor. *Advances in Nucleic Acid Therapeutics*. Cambridge: Royal Society of Chemistry; 2018. p. 32–61.
- Seth PP, Vasquez G, Allerson CA, Berdeja A, Gaus H, Kinberger GA, et al. Synthesis and biophysical evaluation of 2',4'-constrained 2'-O-methoxyethyl nucleic acid analogues. *J Org Chem*. 2010;75(5):1569–1581.
- Seth, P. P., Vasquez, G., Allerson, C. A., Berdeja, A., Gaus, H., Kinberger, G. A., Prakash, T. P., Migawa, M. T., Bhat, B., & Swayze, E. E. (2010). Synthesis and biophysical evaluation of 2',4'-constrained 2'-O-ethyl and 2',4'-constrained 2'-O-methoxyethyl nucleic acid analogues. *Journal of Medicinal Chemistry*, 53(23), 8309–8318.
- Souleimanian, N., Deleavey, G.F., Soifer, H., Wang, S., Tiemann, K., Damha, M.J., & Stein, C.A. (2012). Antisense 2'-deoxy-2'-fluoroarabino nucleic acid (2'F-ANA) oligonucleotides: In vitro gymnotic silencers of gene expression. *Molecular Therapy – Nucleic Acids*.
- Southwell, A.L. et al. (2014) 'In vivo evaluation of candidate allele-specific mutant huntingtin gene silencing antisense oligonucleotides', *Molecular Therapy*, 22(12), pp. 2093–2106.
- Stein CA, Cohen JS. Physicochemical properties of phosphorothioate oligodeoxynucleotides. *Nucleic Acids Res*. 1988;16(7):3209–3221.
- Wahba, L., Amon, J.D., Koshland, D. and Vuica-Ross, M., 2011. RNase H and multiple RNA biogenesis factors cooperate to prevent RNA-DNA hybrids from generating genome instability. *Molecular Cell*, 44(6), pp.978–988.
- Zhang, L., Peritz, A. and Meggers, E., 2005. A simple glycol nucleic acid. *Journal of the American Chemical Society*, 127(12), pp.4174–4175.
- Zhao, B. S., & He, C. (2015). Pseudouridine in a new era of RNA modifications. *Cell Research*, 25, 153–154

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- Adams, D., Gonzalez-Duarte, A., O'Riordan, W.D., Yang, C.C., Ueda, M., Kristen, A.V., Tournay, I., Schmidt, H.H., Coelho, T., Berk, J.L. and Lin, K.P. (2018) Patisiran, an RNAi therapeutic, for hereditary transthyretin amyloidosis. *New England Journal of Medicine*, 379(1), pp.11–21.
- Adams D, Tournay IL, Taylor MS, et al. Efficacy and safety of vutrisiran in hereditary transthyretin-mediated amyloidosis (HELIOS-A). *Amyloid*. 2023;30(2):73–82.
- Alnylam Pharmaceuticals, Inc. (2019) GIVLAARI (givosiran) prescribing information. U.S. Food and Drug Administration. Available at: <https://WWW.FDA.GOV/DRUGSATFDA>
- Ayyar, V.S. and Song, D., 2023. Pharmacokinetics, pharmacodynamics, and translational considerations for siRNA therapeutics. (Review article).
- Ayyar, V.S. and Song, D. (2024) 'Mechanistic pharmacokinetics and pharmacodynamics of GalNAc-siRNA: translational model involving competitive receptor-mediated disposition and RISC-dependent gene silencing applied to givosiran', *Journal of Pharmaceutical Sciences*, 113(1), pp. 176–190.
- Berezna, S.Y., Supekova, L., Schultz, P.G. and Deniz, A.A. (2006) 'siRNA in human cells selectively localizes to target RNA sites', *Proceedings of the National Academy of Sciences*, 103(20), pp. 7682–7687.
- Brown, C.R. et al. (2020) 'Investigating the pharmacodynamic durability of GalNAc-siRNA conjugates', *Nucleic Acids Research*, 48(21), pp. 11827–11844.

- Cuellar, T.L., Barnes, D., Nelson, C., Tanguay, J., Yu, S.-F., Wen, X., Scales, S.J., Gesch, J., Davis, D., van Brabant Smith, A., Leake, D., Vandlen, R. and Siebel, C.W. (2015) 'Systematic evaluation of antibody-mediated siRNA delivery using an industrial platform of THIOMAB-siRNA conjugates', *Nucleic Acids Research*, 43(2), pp. 1189–1203. doi: 10.1093/nar/gku1362.
- Du Rietz, H. et al. (2020) Imaging small molecule-induced endosomal escape of siRNA. *Nature Communications*, 11, 1809.
- Elbashir, S.M., Harborth, J., Lendeckel, W., Yalcin, A., Weber, K. and Tuschl, T. (2001) 'Duplexes of 21-nucleotide RNAs mediate RNA interference in cultured mammalian cells', *Nature*, 411, pp. 494–498.
- European Medicines Agency. Leqvio (inclisiran) EPAR: assessment report. Amsterdam: EMA; 2020.
- FDA (2020) Integrated Review: Oxlumio (lumasiran), NDA 214103. U.S. Food and Drug Administration.
- Fire, A., Xu, S., Montgomery, M.K., Kostas, S.A., Driver, S.E. and Mello, C.C. (1998) 'Potent and specific genetic interference by double-stranded RNA in *Caenorhabditis elegans*', *Nature*, 391(6669), pp. 806–811.
- Fitzgerald K, et al. A highly durable RNAi therapeutic inhibitor of PCSK9. *N Engl J Med*. 2017;376:41–51.
- Gane, E., Lim, Y.-S., Kim, J.B., Visvanathan, K., Sicard, E., Fedorak, R.N., Roberts, S.K., Massetto, B., Ye, X., Huang, Y., et al. (2023) 'Safety and antiviral activity of elebsiran (ALN-HBV02) in patients with chronic hepatitis B: results from a phase 1/2 study', *Journal of Hepatology*, 78(4), pp. 708–717.
- Garrelfs SF, et al. Lumasiran, an RNAi therapeutic for primary hyperoxaluria type 1. *N Engl J Med*. 2021;384:1216–1226.
- Gilleron, J. et al. (2013) Image-based analysis of lipid nanoparticle-mediated siRNA delivery, intracellular trafficking and endosomal escape. *Nature Biotechnology*, 31, pp.638–646.
- Gilleron, J. et al. (2013) 'Image-based analysis of lipid nanoparticle-mediated siRNA delivery', *Nature Biotechnology*, 31, pp. 638–646.
- Habtemariam BA, Karsten V, Attarwala H, et al. Pharmacokinetics and pharmacodynamics of vutrisiran. *Clin Pharmacol Ther*. 2021;110(3):702–713.
- Hall AH, Wan J, Shaughnessy EE, Ramsay Shaw B, Alexander KA. RNA interference using boranophosphate siRNAs: structure–activity relationships. *Nucleic Acids Res*. 2004;32(20):5991–6000.
- Hammond, S.M., 2005. Dicing and slicing: the core machinery of the RNA interference pathway. *FEBS Letters*, 579(26), pp.5822–5829.
- Hannon, G.J. (2002) 'RNA interference', *Nature*, 418(6894), pp. 244–251.
- Hefner, E., Clark, K., Whitman, C., Behlke, M.A., Rose, S.D., Peek, A.S. and Rubio, T. (2008) Increased potency and longevity of gene silencing using validated Dicer substrates. *Journal of Biomolecular Techniques*, 19(4), pp. 231–237.
- Hoy, S.M. (2018) Patisiran: First global approval. *Drugs*, 78(15), pp.1625–1631.
- Janas, M.M., Zlatev, I., Liu, J., Jiang, Y., Barros, S.A., Sutherland, J.E., Davis, W.P., Liu, J., Brown, C.R., Liu, X., Schlegel, M.K., Blair, L., Zhang, X., Das, B., Tran, C., Aluri, K., Li, J., Agarwal, S., Indrakanti, R., Charisse, K., Nair, J.K., Matsuda, S., Rajeev, K.G., Zimmermann, T., Sepp-Lorenzino, L., Xu, Y., Akinc, A., Fitzgerald, K., Vaishnaw, A.K., Smith, P.F., Manoharan, M., Jadhav, V., Wu, J.-T. and Maier, M.A. (2019) 'Safety evaluation of 2'-deoxy-2'-fluoro nucleotides in GalNAc-siRNA conjugates', *Nucleic Acids Research*, 47(7), pp. 3306–3320. doi: 10.1093/nar/gkz140.

- Nair, J.K. et al., 2014. Multivalent N-acetylgalactosamine-conjugated siRNA localizes in hepatocytes and elicits robust RNAi-mediated gene silencing. *Journal of the American Chemical Society*, 136(49), pp.16958–16961.
- Nair JK, et al. Multivalent N-acetylgalactosamine-conjugated siRNA localizes in hepatocytes and elicits robust RNAi. *J Am Chem Soc.* 2014;136:16958–16961.
- Nanna, A.R., Kel'in, A.V., Theile, C., Pierson, J.M., Voo, Z.X., Garg, A., Nair, J.K., Maier, M.A., Fitzgerald, K. and Rader, C. (2020) 'Generation and validation of structurally defined antibody-siRNA conjugates', *Nucleic Acids Research*, 48, pp. 5281–5293.
- Obici L, Ajroud-Driss S, Lin KP, et al. Impact of vutrisiran on quality of life and physical function in hereditary transthyretin-mediated amyloidosis. *Neurol Ther.* 2023;12:1–15.
- Obici L, Berk JL, González-Duarte A, et al. Quality of life outcomes in APOLLO study of patisiran. *Amyloid.* 2020;27(3):153–162.
- Ohrt, T., Muetze, J., Svoboda, P. and Schwill, P. (2006) 'Intracellular localization and routing of siRNA', *Nucleic Acids Research*, 34(5), pp. 1369–1380.
- Raal FJ, et al. Inclisiran for the treatment of heterozygous familial hypercholesterolemia. *N Engl J Med.* 2020;382:1520–1530.
- Ray, K.K., Landmesser, U., Leiter, L.A., Kallend, D., Dufour, R., Karakas, M., Hall, T., Troquay, R.P., Turner, T., Visseren, F.L.J. and Wijngaard, P. (2017) Inclisiran in patients at high cardiovascular risk with elevated LDL cholesterol. *New England Journal of Medicine*, 376(15), pp.1430–1440.
- Schlegel, M.K., Foster, D.J., Kel'in, A.V., Zlatev, I., Bisbe, A., Jayaraman, M., Lackey, J.G., Rajeev, K.G., Charisse, K., Harp, J. and Manoharan, M., 2017. Chirality dependent potency enhancement and structural impact of glycol nucleic acid modification on siRNA. *Journal of the American Chemical Society*, 139(25), pp.8537–8546.
- Schlegel, M.K., Janas, M.M., Jiang, Y., Barry, J.D., Davis, W., Agarwal, S., Berman, D., Brown, C.R., Castoreno, A., LeBlanc, S. and Manoharan, M., 2022. From bench to bedside: improving the clinical safety of GalNAc-siRNA conjugates using seed-pairing destabilization. *Nucleic Acids Research*, 50(12), pp.6656–6670.
- Springer, A.D. and Dowdy, S.F., 2018. GalNAc-siRNA conjugates: leading the way for delivery of RNAi therapeutics. *Nucleic Acid Therapeutics*, 28(3), pp.109–118.
- Østergaard, M.E., Carrer, M., Anderson, B.A., Afetian, M., Bakooshli, M.A., Santos, J.A., Klein, S.K., Capitano, J., Freestone, G.C., Tanowitz, M., Galindo-Murillo, R., Gaus, H.J., Dwyer, C.A., Jackson, M., Jafar-nejad, P., Rigo, F., Seth, P.P., Gaynor, K.U., Stanway, S.J., Urbonas, L., St. Denis, M.A., Pellegrino, S., Bezerra, G.A., Rigby, M., Gowans, E., Van Rietschoten, K., Beswick, P., Chen, L., Skynner, M.J. and Swayze, E.E. (2025) 'Conjugation to a transferrin receptor 1-binding Bicycle peptide enhances ASO and siRNA potency in skeletal and cardiac muscles', *Nucleic Acids Research*, 53, gkaf270. Available at: <https://doi.org/10.1093/nar/gkaf270>.
- Sugo, T., Terada, M., Oikawa, T., Miyata, K., Nishimura, S., Kenjo, E., Ogasawara-Shimizu, M., Makita, Y., Imaichi, S. and Murata, S. (2016) 'Development of antibody-siRNA conjugate targeted to cardiac and skeletal muscles', *Journal of Controlled Release*, 237, pp. 1–13.
- Sutherland, J.E., Hettinger, J.L., Chan, A., Gilbert, J., Warner, G.L. and Davis, W.P. (2020) 'Nonclinical Safety Profile of Revusiran, a 1st-Generation GalNAc-siRNA Conjugate for Treatment of Hereditary Transthyretin-Mediated Amyloidosis', *Nucleic Acid Therapeutics*, 30(1), pp. 33–49. doi: 10.1089/nat.2019.0796.
- U.S. Food and Drug Administration. Pharmacology/Toxicology Review: Inclisiran (NDA 214012). Silver Spring, MD: FDA; 2021.

- Wilson, R.C. and Doudna, J.A. (2013) 'Molecular mechanisms of RNA interference', *Annual Review of Biophysics*, 42, pp. 217–239.
- Wittrup, A. et al. (2015) 'Visualizing lipid-formulated siRNA release from endosomes and target gene knockdown', *Nature Biotechnology*, 33, pp. 870–876.
- Wittrup A, Lieberman J. Knocking down disease: a progress report on siRNA therapeutics. *Nat Rev Genet.* 2015;16(9):543–552.
- Yuan, Y.R., Pei, Y., Chen, H.Y., Tuschl, T., Patel, D.J. (2006) A Potential Protein-RNA Recognition Event along the RISC-Loading Pathway from the Structure of A. aeolicus Argonaute with Externally Bound siRNA. (2006) *Structure* 14: 1557-1565 17027504 Search on PubMedSearch on PubMed Central DOI: <https://doi.org/10.1016/j.str.2006.08.009> Primary Citation Related Structures: 2F8S, 2F8T
- Yuen, M.-F., Locarnini, S., Lim, Y.-S., Strasser, S.I., Sievert, W., Cheng, W., Subramanian, G.M., Myers, R.P., and Gane, E. (2022) 'Elebsiran, an RNA interference therapeutic targeting hepatitis B virus, demonstrates sustained HBsAg reduction in chronic hepatitis B patients', *Hepatology*, 76(5), pp. 1231–1242.
- Zamore, P.D. (2006) 'RNA interference: Big applause for silencing in Stockholm', *Cell*, 127(6), pp. 1083–1086.
- Zimmermann, T.S., Karsten, V., Chan, A., Chiesa, J., Boyce, M., Bettencourt, B.R., Hutabarat, R., Nochur, S., Vaishnav, A. and Gollob, J. (2017) 'Clinical Proof of Concept for a Novel Hepatocyte-Targeting GalNAc-siRNA Conjugate', *Molecular Therapy*, 25(1), pp. 71–78. doi: 10.1016/j.ymthe.2016.10.019.
- mRNA therapeutics, cap biology, UTRs, and vaccines**
- Aartsma-Rus, A., Corey, D. R., and others (2023). The evolving landscape of antisense oligonucleotide therapeutics. *Nat Rev Drug Discov.*, 22, 1–22.  
(Accessed: 9 May 2026).es escape of TAT-fusion proteins. *Nature Medicine*, 10, pp.310–315.
- Anderson, B.R., Muramatsu, H., Jha, B.K., Silverman, R.H., Weissman, D. and Karikó, K. (2011) 'Nucleoside modifications in RNA limit activation of 2'-5'-oligoadenylate synthetase and increase resistance to cleavage by RNase L', *Nucleic Acids Research*, 39(21), pp. 9329–9338.
- Balatsos, N. et al. (2006) 'Inhibition of mRNA deadenylation by the nuclear cap binding complex (CBC)', *Journal of Biological Chemistry*, 281, pp. 4517–4522.
- Banerjee, A.K. (1980) '5'-terminal cap structure in eucaryotic messenger ribonucleic acids', *Microbiological Reviews*, 44(2), pp. 175–205.
- Blakney, A.K., Ip, S. and Geall, A.J. (2021) An update on self-amplifying mRNA vaccine development. *Vaccines*, 9(2), p.97.
- Buxbaum, A.R., Haimovich, G. and Singer, R.H. (2015) 'In the right place at the right time: visualizing and understanding mRNA localization', *Nature Reviews Molecular Cell Biology*, 16(2), pp. 95–109.
- Buxbaum, A.R., Wu, B. and Singer, R.H. (2014) 'Single  $\beta$ -actin mRNA detection in neurons reveals a mechanism for regulating its translatability', *Science*, 343(6169), pp. 419–422.
- Chahal, G.S., Helbig, K.J., Parton, R.G. and Monson, E.A. (2026) The Biology of Endosomal Escape: Strategies for Enhanced Delivery of Therapeutics. *ACS Nano*, 20, pp. 1789–1813.
- Cheng, H. et al. (2006) 'Human mRNA export machinery recruited to the 5' end of mRNA', *Cell*, 127, pp. 1389–1400.

- Cowling, V.H. (2010) 'Regulation of mRNA cap methylation', *Biochemical Journal*, 425, pp. 295–302.
- Daffis, S., Szretter, K.J., Schriewer, J., Li, J., Youn, S., Errett, J., Lin, T.Y., Schneller, S., Züst, R., Dong, H., Thiel, V., Sen, G.C., Fensterl, V., Klimstra, W.B., Pierson, T.C., Buller, R.M.L., Gale, M. Jr. and Diamond, M.S., 2010. 2'-O methylation of the viral mRNA cap evades host restriction by IFIT family members. *Nature*, 468(7322), pp.452–456.
- Dominski, Z. and Kole, R. (1993) Restoration of correct splicing in thalassemic pre-mRNA by antisense oligonucleotides. *Proceedings of the National Academy of Sciences of the USA*, 90(18), pp. 8673–8677.
- Dowdy, S.F. (2023) 'Endosomal escape of RNA therapeutics: How do we solve this rate-limiting problem?', *RNA*, 29, pp. 396–401.
- Dynes, J.L. and Steward, O. (2007) 'Dynamics of bidirectional transport of Arc mRNA in neuronal dendrites', *Journal of Comparative Neurology*, 500(3), pp. 433–447.
- Edery, I., Humbelin, M., Darveau, A., Lee, K.A.W., Milburn, S., Hershey, J.W.B., Trachsel, H. and Sonenberg, N. (1983) 'Involvement of eukaryotic initiation factor 4A in the cap recognition process', *Journal of Biological Chemistry*, 258(18), pp. 11398–11403.
- Fahrni, M.L., Ismail, I.A.-N., Refi, D.M., Almeman, A., Che Yaakob, N., Md Saman, K., Mansor, N.F., Noordin, N. and Babar, Z.-U.-D. (2022) 'Management of COVID-19 vaccines cold chain logistics: a scoping review', *Journal of Pharmaceutical Policy and Practice*, 15, Article 16. doi: 10.1186/s40545-022-00411-5.
- Furuichi, Y. (2015) 'Discovery of m7G-cap in eukaryotic mRNAs', *Proceedings of the Japan Academy, Series B*, 91(8), pp. 394–409.
- Furuichi, Y. and Shatkin, A.J. (2000) 'Viral and cellular mRNA capping: past and prospects', *Advances in Virus Research*, 55, pp. 135–184.
- Furuichi, Y., LaFiandra, A. and Shatkin, A.J. (1977) '5'-Terminal structure and mRNA stability', *Nature*, 266, pp. 235–239.
- Gonatopoulos-Pournatzis, T. et al. (2011) 'RAM/Fam103a1 is required for mRNA cap methylation', *Molecular Cell*, 44, pp. 585–596.
- Gonatopoulos-Pournatzis, T. and Cowling, V.H. (2014) 'Cap-binding complex (CBC)', *Biochemical Journal*, 457, pp. 231–242.
- Grau, M. and Wagner, E. (2024) Strategies and mechanisms for endosomal escape of therapeutic nucleic acids. *Current Opinion in Chemical Biology*, 81, 102506.
- Grudzien-Nogalska, E., Stepinski, J., Jemielity, J., Zuberek, J., Stolarski, R., Rhoads, R.E. and Darzynkiewicz, E. (2007) 'Synthesis of Anti-Reverse Cap Analogs (ARCAs) and their Applications in mRNA Translation and Stability', *Methods in Enzymology*, 431, pp. 203–227.
- Habjan, M., Hubel, P., Lacerda, L., Benda, C., Holze, C., Eberl, C.H., Mann, A., Kindler, E., Gil-Cruz, C., Ziebuhr, J., Thiel, V., Pichlmair, A. and Schneider, W.M., 2013. Sequestration by IFIT1 impairs translation of 2'O-unmethylated capped RNA. *PLoS Pathogens*, 9(10), e1003663.
- Hou X, Zaks T, Langer R, Dong Y. Lipid nanoparticles for mRNA delivery. *Nat Rev Mater*. 2021;6:1078–1094.
- Izaurrealde, E. et al. (1994) 'A nuclear cap binding protein complex involved in pre-mRNA splicing', *Cell*, 78, pp. 657–668.
- Izaurrealde, E. et al. (1995) 'A cap-binding protein complex mediating U snRNA export', *Nature*, 376, pp. 709–712.

- Jemielity, J., Fowler, T., Zuberek, J., Stepinski, J., Lewdorowicz, M., Niedzwiecka, A., Stolarski, R., Darzynkiewicz, E. and Rhoads, R.E. (2003) 'Novel "Anti-Reverse" Cap Analogues with Superior Translational Properties', *RNA*, 9(9), pp. 1108–1122.
- Johnson, B., VanBlargan, L.A., Xu, W., White, J.P., Shan, C., Shi, P.Y., Zhang, R. and Diamond, M.S., 2018. Human IFIT3 modulates IFIT1 RNA binding specificity and protein stability. *Immunity*, 48(3), pp.487–499.
- Jozic, A., Le Roux, C., Kim, J., Berchel, M., Sahel, D.K., Bodi, E.K., Palumbo, M., Vasudevan, A., Murthy, N.T.V., Eygeris, Y., Gautam, M., Bloom, E., Barnes, A.P., Jaffrès, P.-A. and Sahay, G. (2026) 'In vivo endosomal escape assay identifies mechanisms for efficient hepatic LNP delivery', *Nature Biotechnology*.  
<https://doi.org/10.1038/s41587-026-03022-6>
- Karikó, K., Buckstein, M., Ni, H. and Weissman, D. (2005) 'Suppression of RNA recognition by Toll-like receptors: the impact of nucleoside modification and the evolutionary origin of RNA', *Immunity*, 23(2), pp. 165–175. doi: 10.1016/j.immuni.2005.06.008.
- Karikó, K., Muramatsu, H., Welsh, F. A., Ludwig, J., Kato, H., Akira, S., & Weissman, D. (2008). Incorporation of pseudouridine into mRNA yields superior nonimmunogenic vectors with increased translational capacity and biological stability. *Mol Ther*. 16(11), 1833–1840.
- Khong, A., Matheny, T., Jain, S., Mitchell, S.F., Wheeler, J.R. and Parker, R. (2017) 'The stress granule transcriptome reveals principles of mRNA accumulation in stress granules', *Molecular Cell*, 68(4), pp. 808–820.e5.
- Kozak, M. (1979) 'Inability of circular mRNA to attach to eukaryotic ribosomes', *Nature*, 280, pp. 82–85.
- Kozak, M. (1984) 'Compilation and analysis of sequences upstream from the translational start site in eukaryotic mRNAs', *Nucleic Acids Research*, 12(2), pp. 857–872.
- Kozak, M. (1986) 'Point mutations define a sequence flanking the AUG initiator codon that modulates translation by eukaryotic ribosomes', *Cell*, 44(2), pp. 283–292.
- Kozak, M. (1987) 'An analysis of 5'-noncoding sequences from 699 vertebrate messenger RNAs', *Nucleic Acids Research*, 15(20), pp. 8125–8148.
- Kozak, M. (1989) 'The scanning model for translation: an update', *Journal of Cell Biology*, 108(2), pp. 229–241.
- Kozak, M. (1999) 'Initiation of translation in prokaryotes and eukaryotes', *Gene*, 234(2), pp. 187–208
- Kozak, M. (2002) 'Pushing the limits of the scanning mechanism for initiation of translation', *Gene*, 299(1–2),
- Lenasi, T., Peterlin, B. and Barboric, M. (2011) 'Cap-binding protein complex links pre-mRNA capping to transcription elongation and alternative splicing through positive transcription elongation factor b (P-TEFb)', *Journal of Biological Chemistry*, 286, pp. 22758–22768.
- Lewis, J.D. et al. (1996) 'A nuclear cap-binding complex facilitates association of U1 snRNP with the cap-proximal 5' splice site', *Genes & Development*, 10, pp. 1683–1698.
- Liang, X.H. et al. (2016) 'Translation efficiency of mRNAs is increased by antisense oligonucleotides targeting upstream open reading frames', *Nature Biotechnology*, 34, pp. 875–880.
- Liu, H., & Kohane, I. S. (2009). Tissue and process specific microRNA–mRNA co-expression in mammalian development and malignancy. *PLoS ONE*, 4(5), e5436.  
<https://doi.org/10.1371/journal.pone.0005436>

- Morais, P., Adachi, H. and Yu, Y.-T. (2021) The critical contribution of pseudouridine to mRNA COVID-19 vaccines. *Frontiers in Cell and Developmental Biology*, 9, 789427.
- Nobel Prize Outreach AB (2006) The Nobel Prize in Physiology or Medicine 2006. Available at: <https://www.nobelprize.org/prizes/medicine/2006/summary/>
- Park, H.Y., Lim, H., Yoon, Y.J., Follenzi, A., Nwokafor, C., Lopez-Jones, M., Meng, X. and Singer, R.H. (2014) ‘Visualization of dynamics of single endogenous mRNA labeled in live mouse’, *Science*, 343(6169), pp. 422–424.
- Pasquinelli, A.E., Dahlberg, J.E. and Lund, E. (1995) ‘Reverse 5’ Caps in RNAs Made In Vitro by Phage RNA Polymerases’, *RNA*, 1(9), pp. 957–967.
- Polack, F.P., Thomas, S.J., Kitchin, N., Absalon, J., Gurtman, A., Lockhart, S., Perez, J.L., Pérez Marc, G., Moreira, E.D., Zerbini, C. and Bailey, R. (2020) Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine. *New England Journal of Medicine*, 383(27), pp.2603–2615.
- Powell, S.T. (1949) ‘Neutralizing amines in steam condensate systems’, *Industrial & Engineering Chemistry*, 41(10), pp. 2215–2219.
- Ray, B.K., Brendler, T.G., Adya, S., Daniels-McQueen, S., Miller, J.K., Hershey, J.W.B., Grifo, J.A., Merrick, W.C. and Thach, R.E. (1985) ‘Role of mRNA competition in regulating translation: further characterization of mRNA discriminatory initiation factors’, *Proceedings of the National Academy of Sciences USA*, 82(1), pp. 105–109.
- Shatkin, A.J. (1976) ‘Capping of eucaryotic mRNAs’, *Cell*, 9(4), pp. 645–653.
- Shine, J. and Dalgarno, L. (1974) ‘The 3’-terminal sequence of *Escherichia coli* 16S ribosomal RNA: complementarity to nonsense triplets and ribosome binding sites’, *Proceedings of the National Academy of Sciences USA*, 71(4), pp. 1342–1346.
- Sood, P., Krek, A., Zavolan, M., Macino, G., & Rajewsky, N. (2006). Cell-type-specific signatures of microRNAs on target mRNA expression. *Proceedings of the National Academy of Sciences*, 103(8), 2746–2751. <https://doi.org/10.1073/pnas.0511045103>
- Stepinski, J., Waddell, C., Stolarski, R., Darzynkiewicz, E. and Rhoads, R.E. (2001) ‘Synthesis and Properties of mRNAs Containing the Novel “Anti-Reverse” Cap Analogues 7-Methyl(3’-O-Methyl)GpppG and 7-Methyl(3’-Deoxy)GpppG’, *RNA*, 7(10), pp. 1486–1495.
- Tenchov R, Sasso JM, Zhou Q. Lipid nanoparticles—from liposomes to mRNA vaccine delivery. *ACS Nano*. 2021;15(11):16982–17015.
- Tian, B. and Manley, J.L. (2017) ‘Alternative polyadenylation of mRNA precursors’, *Nature Reviews Molecular Cell Biology*, 18, pp. 18–30.
- Topisirovic, I. et al. (2011) ‘Cap and cap-binding proteins in the control of gene expression’, *WIREs RNA*, 2, pp. 277–298.
- Yu, R.Z. et al. (1999) ‘Pharmacokinetics and tissue disposition in monkeys of an antisense oligonucleotide inhibitor of Ha-Ras encapsulated in stealth liposomes’, *Pharmaceutical Research*, 16(8), pp. 1309–1315.
- Züst, R., Cervantes-Barragan, L., Habjan, M., Maier, R., Neuman, B.W., Ziebuhr, J., Szretter, K.J., Baker, S.C., Barchet, W., Diamond, M.S., Siddell, S.G., Ludwig, S., Ludwig, B. and Thiel, V., 2011. Ribose 2’-O-methylation provides a molecular signature for the distinction of self and non-self mRNA dependent on the RNA sensor Mda5. *Nature Immunology*, 12(2), pp.137–143.

## Delivery systems: LNPs, GalNAc, PEG, peptides, antibodies, and endosomal escape

Allen TM, Cullis PR. Liposomal drug delivery systems: from concept to clinical applications. *Adv Drug Deliv Rev*. 2013;65(1):36–48.

- Ashwell, G. and Morell, A.G., 1968. The role of surface carbohydrates in the hepatic recognition and transport of circulating glycoproteins. *Advances in Enzymology*, 41, pp.99–128.
- Beck, A., Goetsch, L., Dumontet, C. and Corvaia, N. (2017) ‘Strategies and challenges for the next generation of antibody–drug conjugates’, *Nature Reviews Drug Discovery*, 16(5), pp. 315–337. doi: 10.1038/nrd.2016.268.
- Belgrad, J., Fakih, H.H. and Khvorova, A. (2024) ‘Nucleic acid therapeutics: Successes, milestones, and upcoming innovation’, *Nucleic Acid Therapeutics*, 34, pp. 52–72.
- Cullis, P.R. and Hope, M.J. (2017) Lipid nanoparticle systems for enabling gene therapies. *Molecular Therapy*, 25(7), pp.1467–1475.
- Drago, J.Z., Modi, S. and Chandarlapaty, S. (2021) ‘Unlocking the potential of antibody–drug conjugates for cancer therapy’, *Nature Reviews Clinical Oncology*, 18, pp. 327–344. doi: 10.1038/s41571-021-00470-8.
- Fishbane S, Singh AK, et al. Peginesatide in patients with anemia undergoing hemodialysis. *N Engl J Med*. 2013;368:307–319.
- Gori, A., Lodigiani, G., Colombarolli, S.G., Bergamaschi, G. and Vitali, A. (2023) ‘Cell penetrating peptides: Classification, mechanisms, methods of study, and applications’, *ChemMedChem*, 18, e202300236.
- Guo, Z., Peng, H., Kang, J. and Sun, D. (2016) ‘Cell-penetrating peptides: Possible transduction mechanisms and therapeutic applications’, *Biomedical Reports*, 4, pp. 528–534.
- Hafez, I.M., Maurer, N. and Cullis, P.R. (2001) On the mechanism whereby cationic lipids promote intracellular delivery of polynucleic acids. *Gene Therapy*, 8, pp.1188–1196.
- Harris JM, Chess RB. Effect of PEGylation on pharmaceuticals. *Nat Rev Drug Discov*. 2003;2(3):214–221.
- He, C. et al. (2021) ‘High-resolution visualization and quantification of nucleic acid-based therapeutics using NanoSIMS’, *Nucleic Acids Research*, 49(1), pp. 1–14.
- Juliano, R.L. (2018) ‘Intracellular trafficking and endosomal release of oligonucleotides: what we know and what we don’t’, *Nucleic Acid Therapeutics*, 28(3), pp. 166–177.
- Juliano, R.L. (2021) Chemical manipulation of the endosome trafficking machinery: Implications for oligonucleotide delivery. *Biomedicines*, 9, 512.
- Junutula, J.R., Raab, H., Clark, S., Bhakta, S., Leipold, D.D., Weir, S., Chen, Y., Simpson, M., Tsai, S.P., Dennis, M.S., Lu, Y., Meng, Y.G., Ng, C., Yang, J., Lee, C.C., Duenas, E., Gorrell, J., Katta, V., Kim, A., McDorman, K., Flagella, K., Venook, R., Ross, S., Spencer, S.D., Lee Wong, W., Lowman, H.B., Vandlen, R., Sliwkowski, M.X., Scheller, R.H., Polakis, P. and Mallet, W. (2008) ‘Site-specific conjugation of a cytotoxic drug to an antibody improves the therapeutic index’, *Nature Biotechnology*, 26, pp. 925–932. doi: 10.1038/nbt.1480.
- Li, M. et al. (2025) ‘Advances in the pharmaceutical development of antibody–oligonucleotide conjugates’, *European Journal of Pharmaceutical Sciences*, 215, 107292.
- Mastrobattista, E. et al. (2002) Functional characterization of an endosome-disruptive peptide and its application in cytosolic delivery. *Journal of Biological Chemistry*, 277, pp.27135–27143.
- Morell, A.G. et al., 1966. Physical and chemical studies on ceruloplasmin. *Journal of Biological Chemistry*, 241, pp.3745–3749.
- Pasut G, Veronese FM. State of the art in PEGylation: the great versatility achieved after forty years of research. *J Control Release*. 2012;161(2):461–472.

- Qassem, S., Breier, D., Naidu, G. S., Hazan-Halevy, I., & Peer, D. (2024). Unlocking the therapeutic potential of locked nucleic acids through lipid nanoparticle delivery. *Molecular Therapy: Nucleic Acids*, 35, 1–12.
- Sahay G, Alakhova DY, Kabanov AV. Endocytosis of nanomedicines. *J Control Release*. 2010;145(3):182–195.
- Shastri, M., Gupta, A., Chandarlapaty, S., Young, M., Powles, T. and Hamilton, E. (2023) ‘Rise of antibody-drug conjugates: the present and future’, *ASCO Educational Book*, 43, e390094. doi: 10.1200/EDBK\_390094.
- Talemi, M.N. et al. (2026) ‘Programmable lipid nanoparticles for RNA therapeutics: design principles and clinical translation’, *Materials Today Bio*.
- Tsuchikama, K. and An, Z. (2018) ‘Antibody-drug conjugates: recent advances in conjugation and linker chemistries’, *Protein & Cell*, 9(1), pp. 33–46. doi: 10.1007/s13238-016-0323-0.
- Veronese FM, Mero A. The impact of PEGylation on biological therapies. *BioDrugs*. 2008;22(5):315–329.
- Veronese FM, Pasut G. PEGylation, successful approach to drug delivery. *Drug Discov Today*. 2005;10(21):1451–1458.
- Winkler, J. (2013) ‘Oligonucleotide conjugates for therapeutic applications’, *Therapeutic Delivery*, 4(7), pp. 791–809.
- Yang, D., Liu, B. and Sha, H. (2025) ‘Advances and prospects of cell-penetrating peptides in tumor immunotherapy’, *Scientific Reports*, 15, 3392.

## Morpholinos and splice-switching therapies

- Aartsma-Rus, A. and Corey, D.R. (2020) ‘The 10th oligonucleotide therapy approved: Golodirsen for Duchenne muscular dystrophy’, *Nucleic Acid Therapeutics*, 30(2), pp. 67–70.
- Assefa, M., Gepfert, A., Zaheer, M., Hum, J.M. and Skinner, B.W. (2024) Casimersen (AMONDYS 45™): An Antisense Oligonucleotide for Duchenne Muscular Dystrophy. *Biomedicines*, 12, 912.
- Avidity Biosciences (2024) Anti-transferrin receptor antibody-PMO conjugates for inducing DMD exon 44 skipping, US Patent US 12,071,621 B2.
- Clemens, P.R. et al. (2020) ‘Safety, tolerability, and efficacy of viltolarsen in boys with Duchenne muscular dystrophy’, *JAMA Neurology*, 77(8), pp. 982–991.
- Desaulniers, J.-P., Hammill, M.L., Georgees, I., Chiu, V.W.-N., Pendergraff, H., Aguti, S., Voutila, J., Hansen, H., Habib, N., Lomonosova, Y. and Koch, T. (2025) Exploring chemically modified short activating RNAs to increase stability against nucleases and enhance gene activation. *Journal of Medicinal Chemistry*, 68, pp. 22650–22664.
- Dulla, K. et al. (2021) ‘Antisense oligonucleotide-based treatment of retinitis pigmentosa caused by USH2A exon 13 mutations’, *Molecular Therapy*, 29(7), pp. 2441–2455.
- FDA (2020) VILTEPSO (viltolarsen) prescribing information.
- FDA (2021) AMONDYS 45 (casimersen) Prescribing Information. U.S. Food and Drug Administration.
- Gilman, H. (1935) ‘Organic compounds of lithium’, *Journal of the American Chemical Society*, 57(9), pp. 1744–1750.
- Gilman, H. (1940) ‘The use of amines in organic synthesis’, *Journal of the American Chemical Society*, 62(6), pp. 1513–1518.
- Hua, Y. et al. (2008) ‘Antisense masking of an hnRNP A1/A2 intronic splicing silencer corrects SMN2 splicing in transgenic mice’, *American Journal of Human Genetics*, 82(4), pp. 834–848.

- Knorr, L. (1887) 'Ueber die Einwirkung von Ammoniak auf Diäthylenoxyd', *Berichte der Deutschen Chemischen Gesellschaft*, 20, pp. 909–918.
- Lim, K.H., Han, Z., Jeon, H.Y., Kach, J., Jing, E., Weyn-Vanhentenryck, S., Downs, M., Corrionero, A., Oh, R., Scharner, J., Venkatesh, A., Ji, S., Liao, G., Ticho, B., Nash, H. and Aznarez, I. (2020) 'Antisense oligonucleotide modulation of non-productive alternative splicing upregulates gene expression', *Nature Communications*, 11, article 3501. Available at: <https://doi.org/10.1038/s41467-020-17093-9>.
- Ling, K. et al. (2025) Enhanced splicing modulation by NMA-modified antisense oligonucleotides. *bioRxiv*.
- Mendell, J.R. et al. (2016) 'Eteplirsen for the treatment of Duchenne muscular dystrophy', *Annals of Neurology*, 79(2), pp. 257–271.
- Reid, R.C. and Sherwood, T.K. (1958) *The Properties of Gases and Liquids*. New York: McGraw-Hill.
- Sidgwick, N.V. (1937) *The Organic Chemistry of Nitrogen*. Oxford: Clarendon Press.
- Stahl, M. et al. (2024) AOC 1044 as a novel therapeutic approach for DMD patients amenable to exon 44 skipping: EXPLORE44 Phase 1/2 healthy volunteer data, MDA Clinical & Scientific Conference poster.
- Summerton, J. (2016) 'History and properties of morpholino antisense oligonucleotides', *Austin Journal of Drug Development and Therapeutics*, 3(1), pp. 1–8.
- Summerton, J. (2017) 'Invention and early history of Morpholinos', in *Methods in Molecular Biology*. New York: Springer, pp. 1–15.
- Summerton, J. and Weller, D. (1997) 'Morpholino antisense oligomers: design, preparation, and properties', *Antisense & Nucleic Acid Drug Development*, 7(3), pp. 187–195.
- Whitmore, F.C. (1937) *Organic Chemistry*. New York: D. Van Nostrand Company.
- Yang, P. et al. (2025) 'Safety and efficacy of ultevursen for the treatment of USH2A-associated retinitis pigmentosa', presented at ARVO 2025.

## Aptamers and RNA localization tools

- Cawte, A.D., Unrau, P.J. and Rueda, D.S. (2020) 'Live cell imaging of single RNA molecules with fluorogenic Mango II arrays', *Nature Communications*, 11, p. 1283.
- Dolgosheina, E.V. et al. (2014) 'RNA Mango aptamer-fluorophore: a bright, high-affinity complex for RNA labeling and tracking', *ACS Chemical Biology*, 9(10), pp. 2412–2420.
- Hinnebusch, A.G. (2011) 'Molecular mechanism of scanning and start codon selection in eukaryotes', *Microbiology and Molecular Biology Reviews*, 75(3), pp. 434–467.
- Keefe, A.D., Pai, S. and Ellington, A. (2010) 'Aptamers as therapeutics', *Nature Reviews Drug Discovery*, 9(7), pp. 537–550.
- Ng, E.W.M. et al. (2006) 'Pegaptanib, a targeted anti-VEGF aptamer for ocular vascular disease', *Nature Reviews Drug Discovery / related structural analyses*.
- Paige, J.S., Wu, K.Y. and Jaffrey, S.R. (2011) 'RNA mimics of green fluorescent protein', *Science*, 333(6042), pp. 642–646.
- Reebye, V., Saetrom, P., Mintz, P.J., Huang, K.W., Swiderski, P., Peng, L., Liu, C., Liu, X., Lindkær-Jensen, S., Zacharoulis, D. and Kostomitsopoulos, N. (2018) Novel RNA oligonucleotide improves liver function and inhibits liver carcinogenesis in vivo. *Hepatology*, 67(2), pp.561–575.
- Ruckman, J. et al. (1998) '2'-Fluoropyrimidine RNA-based aptamers to vascular endothelial growth factor (VEGF165)', *Journal of Biological Chemistry*, 273(32), pp. 20556–20567.
- Williams, A.H., O'Donnell, C., Sejnowski, T.J. and O'Leary, T. (2016) 'Dendritic trafficking faces a traffic jam', *Neuron*, 92(4), pp. 729–740.

## microRNA, saRNA, RNA activation, and RNA editing

- Bartel, D. P. (2018). Metazoan microRNAs. *Cell*, 173(1), 20–51.  
<https://doi.org/10.1016/j.cell.2018.03.006>
- Beg, M.S., Brenner, A.J., Sachdev, J., Borad, M., Kang, Y.K., Stoudemire, J., Smith, S., Bader, A.G. and Kim, S. (2017) Phase I study of MRX34, a liposomal miR-34a mimic, administered twice weekly in patients with advanced solid tumors. *Investigational New Drugs*, 35(2), pp.180–188.
- Cheloufi, S., Dos Santos, C.O., Chong, M.M.W. and Hannon, G.J. (2010) ‘A dicer-independent miRNA biogenesis pathway that requires Ago catalysis’, *Nature*, 465(7298), pp. 584–589.
- Eisenberg, E., & Levanon, E. Y. (2018). A-to-I RNA editing—immune protector and transcriptome diversifier. *Nat Rev Genet.*
- Elmén, J., Lindow, M., Silahatoglu, A., Bak, M., Christensen, M., Lind-Thomsen, A., Hedtjærn, M., Hansen, J. B., Hansen, H. F., Straarup, E. M., McCullagh, K., Kearney, P., Kauppinen, S., and others (2008). LNA-mediated microRNA silencing in non-human primates. *Nature*, 452, 896–899.
- Gao, F.-B. (2010). Context-dependent functions of specific microRNAs in neuronal development. *Neural Development*, 5, 25. <https://doi.org/10.1186/1749-8104-5-25>
- Hauschild, A.-C., Pastrello, C., Ekaputeri, G. K. A., Bethune-Waddell, D., Abovsky, M., Ahmed, Z., Kotlyar, M., Lu, R., & Jurisica, I. (2023). MirDIP 5.2: tissue context annotation and novel microRNA curation. *Nucleic Acids Research*, 51(D1), D217–D225. <https://doi.org/10.1093/nar/gkac1070>
- Hong, D. S., Kang, Y. K., Borad, M., Sachdev, J., Ejadi, S., Lim, H. Y., et al. (2020). Phase 1 study of MRX34, a liposomal miR-34a mimic, in patients with advanced solid tumours. *British Journal of Cancer*, 122, 1630–1637.
- Iyer, S.G., Sohal, I.S. and Kasinski, A.L. (2025) Redesigning miR-34a: structural and chemical advances in the therapeutic development of a miRNA anti-cancer agent. *Biochemical Society Transactions*. <https://DOI.ORG/10.1042/BST20253010>
- Janowski, B.A., Younger, S.T., Hardy, D.B., Ram, R., Huffman, K.E. and Corey, D.R. (2007) ‘Activating gene expression in mammalian cells with promoter-targeted duplex RNAs’, *Nature Chemical Biology*, 3(3), pp. 166–173.
- Janssen, H. L. A., Reesink, H. W., Lawitz, E. J., Zeuzem, S., Rodriguez-Torres, M., Patel, K., van der Meer, A. J., Patick, A. K., Chen, A., Zhou, Y., Persson, R., King, B. D., Kauppinen, S., Levin, A. A., and Hodges, M. R. (2013). Treatment of HCV infection by targeting microRNA. *N Engl J Med.*, 368(18), 1685–1694.
- Krützfeldt, J. et al. (2005) ‘Silencing of microRNAs in vivo with “antagomirs”’, *Nature*, 438, pp. 685–689.
- Lanford, R. E., et al. (2010). Therapeutic silencing of microRNA-122 in primates with chronic hepatitis C virus infection. *Science*, 327(5962), 198–201.
- Li, L.-C., et al. (2025). RNAa: Mechanisms, therapeutic potential, and clinical progress. *Molecular Therapy — Nucleic Acids*. <https://doi.org/10.1016/j.omtn.2025.102468>
- Neben, S. (2016–2018). RG-101 and miR-122 Antagonist Program Updates. TIDES Conference presentations.
- Nishikura, K. (2010). Functions and regulation of RNA editing by ADAR deaminases. *Annu Rev Biochem*
- Parashar, D., Geethadevi, A., Aure, M.R., Mishra, J., George, J., Chen, C., Mishra, M.K., Tahiri, A., Zhao, W., Nair, B. and et al. (2019) ‘miR551b-3p activates an oncostatin signaling module for the progression of triple-negative breast cancer’, *Cell Reports*, 29(13), pp. 4389–4406.e10.

- Parashar, D., Geethadevi, A., McGuire, M.H., Krishnan, V., Wang, L., Sahin, A.A., Lopez-Berestein, G., Sood, A.K. and Ram, P.T. (2016) 'Direct upregulation of STAT3 by microRNA-551b-3p deregulates growth and metastasis of ovarian cancer', *Cell Reports*, 15(7), pp. 1493–1504.
- Place, R. F., Li, L.-C., Pookot, D., Noonan, E. J., & Dahiya, R. (2008). MicroRNA-373 induces expression of genes with complementary promoter sequences. *Proceedings of the National Academy of Sciences*, 105(5), 1608–1613.  
<https://doi.org/10.1073/pnas.0707594105>
- Rainaldi, J., Mali, P. and Nourredine, S. (2025) Emerging clinical applications of ADAR-based RNA editing. *Stem Cells Translational Medicine*, 14(5), szaf016.
- Rishik, S., Hirsch, P., Grandke, F., Fehlmann, T., & Keller, A. (2025). miRNATissueAtlas 2025: an update to the uniformly processed and annotated human and mouse non-coding RNA tissue atlas. *Nucleic Acids Research*, 53(D1), D129–D137.  
<https://doi.org/10.1093/nar/gkae1036>
- Shang, R., Lee, S., Senavirathne, G., & Lai, E. C. (2023). microRNAs in action: biogenesis, function and regulation. *Nature Reviews Genetics*, 24, 816–833.  
<https://doi.org/10.1038/s41576-023-00611-y>
- Tsang, J. S., Ebert, M. S., & van Oudenaarden, A. (2010). Tissue-specific target analysis of disease-associated microRNAs in human signaling pathways. *PLoS ONE*, 5(6), e11154. <https://doi.org/10.1371/journal.pone.0011154>
- van der Ree, M. H., de Vree, J. M. L., Stelma, F., Willemse, S., van der Valk, M., Rietdijk, S., Molenkamp, R., Schinkel, J., van Nuenen, A. C., Beuers, U., Hadi, S., Harbers, M., van der Veer, E., Liu, K., Grundy, J., Patick, A. K., Pavlicek, A., Blem, J., Huang, M., Grint, P., Neben, S., Gibson, N. W., Kootstra, N. A., & Reesink, H. W. (2017). Safety, tolerability, and antiviral effect of RG-101 in patients with chronic hepatitis C: A phase 1B, double-blind, randomised controlled trial. *The Lancet*, 389(10070), 709–717. [https://doi.org/10.1016/S0140-6736\(16\)31715-9](https://doi.org/10.1016/S0140-6736(16)31715-9)
- van der Ree, M.H., van der Meer, A.J., van Nuenen, A.C., de Bruijne, J., Ottosen, S., Janssen, H.L.A. and Reesink, H.W. (2017) Miravirsin, an oligonucleotide targeting miR-122, shows antiviral activity in chronic hepatitis C patients. *Science Translational Medicine*, 9(395), eaal3468.
- Vasudevan, S., Tong, Y., & Steitz, J. A. (2007). Switching from repression to activation: MicroRNAs can up-regulate translation. *Science*, 318(5858), 1931–1934.  
<https://doi.org/10.1126/science.1149460>
- Wei, Y., Li, L., Wang, D., Zhang, C.Y. and Zen, K. (2014) 'Importin 8 regulates the transport of mature microRNAs into the cell nucleus', *Journal of Biological Chemistry*, 289(15), pp. 10270–10275.
- Yang, J.S., Maurin, T., Robine, N., Rasmussen, K.D., Jeffrey, K.L., Chandwani, R., Papapetrou, E.P., Sadelain, M., O'Carroll, D. and Lai, E.C. (2010) 'Conserved vertebrate mir-451 provides a platform for Dicer-independent, Ago2-mediated microRNA
- Zhang, L., Liao, Y., & Tang, L. (2019). MicroRNA-34 family: a potential tumor suppressor and therapeutic candidate in cancer. *Journal of Experimental & Clinical Cancer Research*
- Zhu, W., Yang, L., & Du, Z. (2011). MicroRNA regulation and tissue-specific protein interaction network. *PLoS ONE*, 6(9), e25394.  
<https://doi.org/10.1371/journal.pone.0025394>

## CNS, intrathecal delivery, and gene therapy vectors

- Baltanás, F.C., Zarich, N., Rojas-Cabañeros, J.M. and Santos, E. (2020) ‘SOS GEFs in health and disease’, *BBA - Reviews on Cancer*, 1874, 188445.
- Bier, A. (1899) ‘Experiments regarding the cocainization of the spinal cord’, *\*Classical Articles in Orthopaedics\**, 2(4), pp. 196–203.
- Carroll, J.B. et al. (2011) ‘Potent and selective antisense oligonucleotides targeting single-nucleotide polymorphisms in the Huntington disease gene/allele-specific silencing of mutant huntingtin’, *Molecular Therapy*, 19(12), pp. 2178–2185.
- Chiriboga, C.A. et al. (2016) ‘Results from a phase 1 study of nusinersen in children with spinal muscular atrophy’, *Neurology*, 86(10), pp. 890–897.
- Evers, M.M., Miniarikova, J., Juhas, S., Vallès, A., Bohuslavova, B., Juhasova, J., Skalnikova, H.K., Vodicka, P., Valekova, I., Brouwers, C.C., et al. (2018) ‘AAV5-miHTT gene therapy demonstrates broad distribution and strong human mutant huntingtin lowering in a Huntington’s disease minipig model’, *Molecular Therapy*, 26(9), pp. 2163–2177.
- Finkel, R.S., Mercuri, E., Darras, B.T., Connolly, A.M., Kuntz, N.L., Kirschner, J., Chiriboga, C.A., Saito, K., Servais, L., Tizzano, E. and Topaloglu, H. (2017) Nusinersen versus sham control in infantile-onset spinal muscular atrophy. *New England Journal of Medicine*, 377(18), pp.1723–1732.
- Gaus HJ, Gupta R, Chappell AE, Østergaard ME, Swayze EE, Seth PP. Characterization of the interactions of chemically modified therapeutic nucleic acids with plasma proteins using fluorescence polarization. *Nucleic Acids Res.* 2019;47(2):1110–1122.
- Guimaraes, G., Yuan, L. and Li, P. (2022) Antisense Oligonucleotide In Vitro Protein Binding Determination in Plasma, Brain, and Cerebral Spinal Fluid Using Hybridization LC-MS/MS. *Drug Metabolism and Disposition*, 50, pp. 268–276.
- Iqbal, U., Hwang, R.W. and Costain, W.J. (2026) ‘Recent developments in lipid nanoparticle-mediated delivery of biotherapeutics and gene therapy across the blood–brain barrier’, *BioDrugs*, 40, pp. 407–431. doi:10.1007/s40259-026-00782-0.
- Kaplitt, M.G., Leone, P., Samulski, R.J., Xiao, X., Pfaff, D.W., O’Malley, K.L. and During, M.J., 1994. Long-term gene expression and phenotypic correction using adeno-associated virus vectors in the mammalian brain. *Nature Genetics*, 8(2), pp.148–154.
- Keys, T.E. (1996) *\*The History of Surgical Anesthesia\**. New York: Dover Publications.
- Kim, J. et al. (2019) ‘Patient-customized oligonucleotide therapy for a rare genetic disease’, *New England Journal of Medicine*, 381, pp. 1644–1652.
- Leone, P., Shera, D., McPhee, S.W.J., Francis, J.S., Kolodny, E.H., Bilaniuk, L.T., Wang, D.J., Assadi, M., Goldfarb, O., Goldman, H.W. and Freese, A., 2012. Long-term follow-up after gene therapy for Canavan disease. *Science Translational Medicine*, 4(165), 165ra163.
- Levin, A.A. (2019) ‘Treating disease at the RNA level with oligonucleotides’, *New England Journal of Medicine*, 380(1), pp. 57–70.
- Li, L.-C., Okino, S.T., Zhao, H., Pookot, D., Place, R.F., Urakami, S., Enokida, H. and Dahiya, R. (2006) ‘Small dsRNAs induce transcriptional activation in human cells’, *Proceedings of the National Academy of Sciences*, 103(46), pp. 17337–17342.
- Lykken, E.A., Shyng, C., Edwards, R.J., Rozenberg, A. and Gray, S.J. (2018) ‘Recent progress and considerations for AAV gene therapies targeting the central nervous system’, *Journal of Neurodevelopmental Disorders*, 10, article 16.
- McCown, T.J., Xiao, X., Li, J., Breese, G.R. and Samulski, R.J., 1996. Differential and persistent expression patterns of CNS gene transfer by an adeno-associated virus vector. *Brain Research*, 713(1–2), pp.99–107.

- Meijboom, K.E., Wood, M.J.A. and McClorey, G. (2017) Splice-switching therapy for spinal muscular atrophy. *Genes*, 8(6), 161.
- Miller, T.M., Cudkowicz, M.E., Genge, A., Shaw, P.J., Sobue, G., Bucelli, R.C., Chio, A., Van Damme, P., Ludolph, A.C., Glass, J.D. and collaborators, 2022. Trial of antisense oligonucleotide tofersen for SOD1 ALS. *New England Journal of Medicine*, 387(12), pp.1099–1110.
- Morton, W.T.G. (1847) ‘Statement of the discovery of the anaesthetic properties of ether’, *American Journal of the Medical Sciences*, 13(26), pp. 37–43.
- Mussche, S., Devreese, B., Nagabhushan Kalburgi, S., Bachaboina, L., Fox, M.A., Shihabuddin, L.S., Wilson, J.M., Sendera, T.J. and Gray, S.J., 2013. Restoration of cytoskeleton homeostasis after gigaxonin gene transfer for giant axonal neuropathy. *Human Gene Therapy*, 24(2), pp.209–219.
- Prakash, V., Moore, M. and Yáñez-Muñoz, R.J. (2016) ‘Current progress in therapeutic gene editing for monogenic diseases’, *Molecular Therapy*, 24(3), pp. 465–474.

### PK/PD, biodistribution, toxicology, and safety

- Abuchowski A, van Es T, Palczuk NC, Davis FF. Alteration of immunological properties of bovine serum albumin by covalent attachment of polyethylene glycol. *J Biol Chem*. 1977;252(11):3578–3581.
- Alconcel SNS, Baas AS, Maynard HD. FDA-approved poly(ethylene glycol)–protein conjugate drugs. *Polym Chem*. 2011;2(7):1442–1448.
- Bennett, C.F. and Swayze, E.E. (2010) ‘RNA targeting therapeutics: molecular mechanisms of antisense oligonucleotides as a therapeutic platform’, *Annual Review of Pharmacology and Toxicology*, 50, pp. 259–293.
- Burel, S. A., et al. (2016). Unique toxicity profiles of locked nucleic acid (LNA) antisense oligonucleotides. *Nucleic Acid Therapeutics*, 26(4), 199–213.
- CADTH (2020) Clinical Review Report: Inotersen (Tegsedi). Canadian Agency for Drugs and Technologies in Health
- ClinicalTrials.gov. Study of AZD5312 in metastatic castration-resistant prostate cancer. Bethesda, MD: U.S. National Library of Medicine; 2025. Available from: <https://clinicaltrials.gov>.
- Crouch, R.J. and Dirksen, M.-L. (1982) ‘Ribonucleases H’, in *Nucleases*. Cold Spring Harbor Laboratory, pp. 211–241.
- Dowdy, S.F., 2017. Overcoming cellular barriers for RNA therapeutics. *Nature Biotechnology*, 35(3), pp.222–229.
- Faulds, D. and Heel, R.C. (1990) Ganciclovir: A review of its antiviral activity, pharmacokinetic properties and therapeutic use. *Drugs*, 39(4), pp. 597–638.
- FDA (2017) Ganciclovir Injection Label. U.S. Food and Drug Administration. Available at: [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2017/2093471bl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2017/2093471bl.pdf)
- FDA (2026) Spinraza (nusinersen) prescribing information. U.S. Food and Drug Administration.
- Galindo-Murillo et al. (2024). Comparative MD analysis of ASO chemistries.
- Geary, R.S., Henry, S.P. and Grillone, L.R. (2002) Fomivirsen: Clinical pharmacology and potential drug interactions. *Clinical Pharmacokinetics*, 41(4), pp. 255–260.
- Geary, R.S., Norris, D., Yu, R. and Bennett, C.F. (2015) ‘Pharmacokinetics, biodistribution and cell uptake of antisense oligonucleotides’, *Advanced Drug Delivery Reviews*, 87, pp. 46–51.
- Giacomini, K. M., Huang, S. M., Tweedie, D. J., Benet, L. Z., Brouwer, K. L. R., Chu, X., Dahlin, A., Evers, R., Fischer, V., Hillgren, K. M., et al. (2010). Membrane transporters in drug development. *Nat Rev Drug Discov.*, 9(3), 215–236.

- Girach, A. et al. (2022) 'RNA-based therapies in inherited retinal diseases', *Therapeutic Advances in Ophthalmology*, 14, pp. 1–20.  
<https://doi.org/10.1177/25158414221134602>
- Hinz, B., Brune, K. and Pahl, A. (2000) Cyclooxygenase-2 expression in different tissues and the effects of nonsteroidal anti-inflammatory drugs. *Biochemical Pharmacology*, 59(9), pp. 1117–1123.
- Jo, S.J., Chae, S.U., Lee, C.B. and Bae, S.K. (2023) Clinical Pharmacokinetics of Approved RNA Therapeutics. *International Journal of Molecular Sciences*, 24, 746.
- König, J., Nies, A. T., Cui, Y., Leier, I., & Keppler, D. (1999). A novel human canalicular multispecific organic anion transporter (MRP2). *Journal of Clinical Investigation*, 104(2), 167–174.
- Khvorova, A., & Watts, J. K. (2017). The chemical evolution of oligonucleotide therapies of clinical utility. *Nature Biotechnology*, 35, 238–248.
- Lee, Y.C. and Lee, R.T., 2008. Interactions of oligosaccharides and glycoproteins with hepatic carbohydrate receptors. In: *Carbohydrates in Chemistry and Biology*. Wiley-VCH.
- Maquat, L.E., Tarn, W.-Y. and Isken, O. (2010) 'The pioneer round of translation: features and functions', *Cell*, 142, pp. 368–374.
- McCloskey, A. et al. (2012) 'hnRNP C tetramer measures RNA length to classify RNA polymerase II transcripts for export', *Science*, 335, pp. 1643–1646.
- Naganuma, T., Nakagawa, S., Tanigawa, A., Sasaki, Y.F., Goshima, N. and Hirose, T. (2012) 'Alternative 3'-end processing of long noncoding RNA initiates construction of nuclear paraspeckles', *EMBO Journal*, 31(20), pp. 4020–4034. Available at: <https://doi.org/10.1038/emboj.2012.251>
- Nowotny, M., Cerritelli, S.M., Ghirlando, R., Gaidamakov, S.A., Crouch, R.J., Yang, W. (2008) *EMBO J* 27: 1172-1181 PubMed: 18337749 DOI: <https://DOI.ORG/10.1038/EMBOJ.2008.44>
- Prakash TP, Siwkowski A, Allerson CR, Vasquez G, Lee S, Wancewicz EV, et al. Short antisense oligonucleotides with conformationally restricted nucleoside analogues show improved potency without increased toxicity. *J Med Chem*. 2009;52(1):10–13.
- Rüger, J., Ioannou, S., Castanotto, D. and Stein, C.A., 2020. Oligonucleotides to the (Gene) Rescue: FDA approvals 2017–2019. *Trends in Pharmacological Sciences*, 41(1), pp.27–41.
- Srinivasan, S., Torres, A. G., & Ribas de Pouplana, L. (2021). Inosine in Biology and Disease. *Genes*.
- Østergaard ME, De Hoyos CL, Wan WB, Shen W, Low A, Berdeja A, et al. Understanding the effect of controlling phosphorothioate chirality in the DNA gap on the potency and safety of gapmer antisense oligonucleotides. *Nucleic Acids Res*. 2020;48(4):1691–1700.
- Thomas, M., White, R.L. and Davis, R.W. (1976) 'Hybridization of RNA to double-stranded DNA: formation of R-loops', *Proceedings of the National Academy of Sciences USA*, 73, pp. 2294–2298.
- Vane, J.R. and Botting, R.M. (2003) The mechanism of action of aspirin. *Thrombosis Research*, 110(5–6), pp. 255–258.
- Wan, W. B., & Seth, P. P. (2016). The medicinal chemistry of therapeutic oligonucleotides. *Journal of Medicinal Chemistry*, 59(21), 9645–9667.
- Warner, T.D. and Mitchell, J.A. (2004) Cyclooxygenases: new forms, new inhibitors, and lessons from the clinic. *FASEB Journal*, 18(7), pp. 790–804.
- Warner, T.D., Giuliano, F., Vojnovic, I., Bukasa, A., Mitchell, J.A. and Vane, J.R. (1999) Nonsteroid drug selectivities for cyclooxygenase-1 rather than cyclooxygenase-2 are

- associated with human gastrointestinal toxicity. *Proceedings of the National Academy of Sciences USA*, 96(13), pp. 7563–7568.
- Watts, J.K. and Corey, D.R. (2012) ‘Silencing disease genes in the laboratory and the clinic’, *Journal of Pathology*, 226(2), pp. 365–379.
- Webster LR, et al. Randomized clinical trial of naloxegol for opioid-induced constipation. *J Pharmacol Exp Ther*. 2013;347(2):408–416.
- Yang, W. (2011) Nucleases: diversity of structure, function and mechanism.
- Zanardi, T. A., Kim, T.-W., Shen, L., Serota, D., Papagiannis, C., Park, S.-Y., Kim, Y., & Henry, S. P. (2018). Chronic toxicity assessment of 2'-O-methoxyethyl antisense oligonucleotides in mice. *Nucleic Acid Therapeutics*, 28(4), 233–240.
- AstraZeneca. 2024. AZD5312 clinical development updates. AstraZeneca internal and public pipeline reports.
- Belaïdi, A.A., Furtado, D.L., Alves, F., Nisbet, R.M., Ayton, S. and Bush, A. (2025) Engineering mRNA-LNP Medicines for the Ageing Brain: Opportunities and Challenges for Neurodegenerative diseases. Authorea preprint. DOI: 10.22541/au.176004563.31962211/v1.
- Bicycle Therapeutics (2021) ‘Bicycle Therapeutics and Ionis Pharmaceuticals announce collaboration to develop targeted oligonucleotide therapeutics’. Available at: [insert URL] (Accessed: 26 May 2026).
- Brown, C.R. et al. (2020) [keep your existing full Brown reference here].
- de Bono, J., et al. 2023. “Targeting androgen receptor signaling with antisense oligonucleotides in prostate cancer.” *Journal of Clinical Oncology* (or relevant conference proceeding).
- DNase I description (no author listed) Deoxyribonuclease I [EC 3.1.21.1].
- Drygin, D. (2022). Lessons Learned from Preclinical and Clinical Development of microRNA Therapies. *RNA Leaders World Congress*, Basel.
- Gene Tools, LLC (n.d.) History, production and properties of Morpholinos. Available at: [https://www.gene-tools.com/history\\_production\\_and\\_properties](https://www.gene-tools.com/history_production_and_properties)
- GLPBio (2026) Acetaminophen Product Information. Available at: <https://www.glpbio.com/acetaminophen.html>
- Ionis Pharmaceuticals (2021) ‘Ionis and Bicycle Therapeutics announce collaboration to develop targeted oligonucleotide therapeutics’. Available at: [insert URL] (Accessed: 26 May 2026).
- Kang, H., et al. (2024) First-in-human study of RAG-01, a p21-targeting saRNA, in non-muscle invasive bladder cancer. \*\*[Journal/Conference not yet specified – early clinical report / trial NCT06351904]\*.
- Mayo Clinic (n.d.) Inotersen (subcutaneous route) description and precautions. Available at: <https://WWW.MAYOCLINIC.ORG> (31 March 2026).
- MedChemExpress (2026) Aspirin Product Information. Available at: <https://www.medchemexpress.com/aspirin.html>
- MedChemExpress (2026) Ibuprofen Product Information. Available at: <https://www.medchemexpress.com/ibuprofen.html>
- Santaris Pharma A/S (2012). Phase 2a clinical study of miravirsin (SPC3649) in treatment-naïve patients with chronic hepatitis C virus infection. Clinical study reports and summaries.
- TribioScience (2023) Imatinib Technical Data Sheet. Available at: [https://tribioscience.com/wp-content/uploads/2023/05/TBI1060\\_Imatinib\\_V02\\_2023.pdf](https://tribioscience.com/wp-content/uploads/2023/05/TBI1060_Imatinib_V02_2023.pdf)

Wadia, J.S., Stan, R.V. and Dowdy, S.F. (2004) Transducible TAT-HA fusogenic peptide enhance BenchChem (2026) Application Notes and Protocols for Determining Fomivirsen Sodium IC50 in Various Cell Lines. Available at: [https://pdf.benchchem.com/10832/Application\\_Notes\\_and\\_Protocols\\_for\\_Determining\\_Fomivirsen\\_Sodium\\_IC50\\_in\\_Various\\_Cell\\_Lines.pdf](https://pdf.benchchem.com/10832/Application_Notes_and_Protocols_for_Determining_Fomivirsen_Sodium_IC50_in_Various_Cell_Lines.pdf)

## PNAs

- Aunins, T.R. et al. (2020) 'Transcriptome-based design of antisense inhibitors potentiates carbapenem efficacy in CRE *Escherichia coli*', *Proceedings of the National Academy of Sciences*, 117, pp. 30699–30709.
- Datta, B. and Armitage, B.A. (2001) 'Hybridization of PNA to structured DNA targets: quadruplex invasion and the overhang effect', *Journal of the American Chemical Society*, 123, pp. 9612–9619.
- Datta, B., Schmitt, C. and Armitage, B.A. (2003) 'Formation of a PNA2–DNA2 hybrid quadruplex', *Journal of the American Chemical Society*, 125, pp. 4111–4118.
- Demidov, V.V. et al. (1994) 'Stability of peptide nucleic acids in human serum and cellular extracts', *Biochemical Pharmacology*, 48, pp. 1310–1313.
- Egholm, M. et al. (1993) 'PNA hybridizes to complementary oligonucleotides obeying the Watson-Crick hydrogen-bonding rules', *Nature*, 365, pp. 566–568.
- El-Fateh, M., Chatterjee, A. and Zhao, X. (2024) 'A systematic review of peptide nucleic acids (PNAs) with antibacterial activities: efficacy, potential and challenges', *International Journal of Antimicrobial Agents*, 63, 107083.
- Eller, K.A. et al. (2021) 'Facile accelerated specific therapeutic (FAST) platform develops antisense therapies to counter multidrug-resistant bacteria', *Communications Biology*, 4, 331.
- Good, L. et al. (2001) 'Bactericidal antisense effects of peptide-PNA conjugates', *Nature Biotechnology*, 19, pp. 360–364.
- Nielsen, P.E. et al. (1991) 'Sequence-selective recognition of DNA by strand displacement with a thymine-substituted polyamide', *Science*, 254, pp. 1497–1500.
- Rapireddy, S. et al. (2007) 'Strand invasion of mixed-sequence B-DNA by acridine-linked,  $\gamma$ -peptide nucleic acid ( $\gamma$ -PNA)', *Journal of the American Chemical Society*, 129, pp. 15596–15600.
- Smith, J.O., Olson, D.A. and Armitage, B.A. (1999) 'Molecular recognition of PNA-containing hybrids: spontaneous assembly of helical cyanine dye aggregates on PNA templates', *Journal of the American Chemical Society*, 121, pp. 2686–2695.
- Brazil, R. (2023) 'Peptide nucleic acids promise new therapeutics and gene editing tools', *ACS Central Science*, 9, pp. 3–6.
- ClinicalTrials.gov (2022) 'A Proof-of-Concept Study Evaluating EOM613 in COVID-19 Infected Patients With Severe Symptoms', NCT05212532.
- ClinicalTrials.gov (2023) 'A 2-stage, Phase 2a Study to Evaluate the Efficacy, Safety, and Tolerability of OLP-1002 Subcutaneous Injections for Reducing Moderate to Severe Pain Due to Osteoarthritis in a Hip and/or Knee Joint', NCT05216341.
- Crooke, S.T. et al. (2020) 'Phosphorothioate modified oligonucleotide–protein interactions', *Nucleic Acids Research*, 48(10), pp. 5235–5253.
- FDA (2016) EXONDYS 51 (etepirsen) injection: prescribing information. US Food and Drug Administration.
- Heo, Y.A. et al. (2024) 'Characterization of nonclinical drug metabolism and pharmacokinetic properties of phosphorodiamidate morpholino oligomers', *Drug Metabolism and Disposition*.

- Liu, X. et al. (2025) ‘Advances in peptide nucleic acid for targeting RNA and genomic DNA’, *Trends in Biotechnology*.
- McMahon, B.M. et al. (2002) ‘Pharmacokinetics and tissue distribution of a peptide nucleic acid after intravenous administration’, *Antisense and Nucleic Acid Drug Development*, 12(2), pp. 65–70.
- OliPass Corporation (2023a) Nav1.7 selective inhibitor OLP-1002 shows strong efficacy and long therapeutic duration according to interim findings from a Phase 2a study in OA patients. PR Newswire/BioSpace, 16 March.
- OliPass Corporation (2023b) OliPass discloses painful but hilarious clinical findings from a Phase 2a trial in OA patients with Nav1.7 selective inhibitor OLP-1002. PR Newswire, 13 November.
- Spence, H., Waldron, F.M., Saleeb, R.S., Brown, A.-L., Rifai, O.M., Gilodi, M., Read, F., Roberts, K., Milne, G., Wilkinson, D., O’Shaughnessy, J., Pastore, A., Fratta, P., Schneider, N., Tartaglia, G.G., Zacco, E., Horrocks, M.H. and Gregory, J.M. (2024) ‘RNA aptamer reveals nuclear TDP-43 pathology is an early aggregation event that coincides with STMN-2 cryptic splicing and precedes clinical manifestation in ALS’, *Acta Neuropathologica*, 147(1), article 50. doi: 10.1007/s00401-024-02705-1.
- Strack, R.L., Disney, M.D. and Jaffrey, S.R. (2013) ‘A superfolding Spinach2 reveals the dynamic nature of trinucleotide repeat-containing RNA’, *Nature Methods*, 10(12), pp. 1219–1224.

## ADAR

- Bass, B.L. and Weintraub, H. (1987) ‘A developmentally regulated activity that unwinds RNA duplexes’, *\*Cell\**, 48, pp. 607–613.
- Bass, B.L. and Weintraub, H. (1988) ‘An unwinding activity that covalently modifies its double-stranded RNA substrate’, *\*Cell\**, 55, pp. 1089–1098.
- Cameau, E., Zhang, P., Ip, S., Mathiasson, L. and Stenklo, K. (2022) ‘Process & analytical insights for GMP manufacturing of mRNA lipid nanoparticles’, *\*Cell & Gene Therapy Insights\**, 8(4), pp. 621–635.
- Higuchi, M. et al. (1993) ‘RNA editing of AMPA receptor subunit GluR-B: a base-paired intron-exon structure determines position and efficiency’, *\*Cell\**, 75, pp. 1361–1370.
- Jeffs, L., Ho, L. and Kapatrala, R. (2024) ‘GMP manufacturing of mRNA-LNP drug products: navigating scale, process controls, and personalized medicine’, *\*BioProcess International Special Report: Navigating GMP Manufacturing of mRNA-Lipid Nanoparticles\**, 22(1–2)si, pp. 4–7.
- Johnson, A., Ho, L. and Kapatrala, R. (2024) ‘Navigating regulatory complexities in the clinical and commercial manufacturing of lipid nanoparticles’, *\*BioProcess International Special Report: Navigating GMP Manufacturing of mRNA-Lipid Nanoparticles\**, 22(1–2)si, pp. 2–4.
- Kim, U. et al. (1994) ‘Molecular cloning of cDNA for double-stranded RNA adenosine deaminase, a candidate enzyme for nuclear RNA editing’, *\*Proceedings of the National Academy of Sciences USA\**, 91, pp. 11457–11461.
- Levanon, E.Y. et al. (2004) ‘Systematic identification of abundant A-to-I editing sites in the human transcriptome’, *\*Nature Biotechnology\**, 22, pp. 1001–1005.
- Maas, S., Gerber, A.P. and Rich, A. (1999) ‘Identification and characterization of a human tRNA-specific adenosine deaminase related to the ADAR family of pre-mRNA editing enzymes’, *\*Proceedings of the National Academy of Sciences USA\**, 96, pp. 8895–8900.
- Melcher, T. et al. (1996) ‘A mammalian RNA editing enzyme’, *\*Nature\**, 379, pp. 460–464.

- MilliporeSigma (2026) \*Regulatory Considerations for Excipients Used in Lipid Nanoparticles\*. White paper. Burlington, MA: Merck KGaA / MilliporeSigma.
- Polson, A.G. et al. (1991) 'The mechanism of adenosine to inosine conversion by the double-stranded RNA unwinding/modifying activity: a high-performance liquid chromatography-mass spectrometry analysis', *\*Biochemistry\**, 30, pp. 11507–11514.
- Stephens, O. (2022) \*Developing Drug Products Containing Nanomaterials: CDER CMC Perspective\*. FDA NanoDay Symposium presentation. Silver Spring, MD: U.S. Food and Drug Administration, Center for Drug Evaluation and Research.
- Wagner, R.W. et al. (1989) 'A double-stranded RNA unwinding activity introduces structural alterations by means of adenosine to inosine conversions in mammalian cells and *Xenopus* eggs', *\*Proceedings of the National Academy of Sciences USA\**, 86, pp. 2647–2651.
- Wright, A.L. et al. (2023) 'The Q/R editing site of AMPA receptor GluA2 subunit acts as an epigenetic switch regulating dendritic spines, neurodegeneration and cognitive deficits in Alzheimer's disease', *\*Molecular Neurodegeneration\**, 18, article 65.

## LNP

- Alnylam Pharmaceuticals (2019) \*Alnylam announces approval of GIVLAARI™ (givosiran) by the U.S. Food and Drug Administration\*. Cambridge, MA: Alnylam Pharmaceuticals.
- Anderson, K.P. et al. (1998) 'Human cytomegalovirus mutant with sequence-dependent resistance to the phosphorothioate oligonucleotide fomivirsen', *Antimicrobial Agents and Chemotherapy*, 42(4), pp. 971–973.
- Ashwell, G. and Morell, A.G. (1974) 'The role of surface carbohydrates in the hepatic recognition and transport of circulating glycoproteins', *\*Advances in Enzymology and Related Areas of Molecular Biology\**, 41, pp. 99–128.
- Benizri, S., Gissot, A., Martin, A., Vialet, B., Grinstaff, M.W. and Barthélémy, P. (2020) 'Bioconjugated oligonucleotides: recent developments and therapeutic applications', *Bioconjugate Chemistry*, 31(2), pp. 366–383.
- Bennett, C.F., Baker, B.F., Pham, N., Swayze, E. and Geary, R.S. (2017) 'Pharmacology of antisense drugs', *Annual Review of Pharmacology and Toxicology*, 57, pp. 81–105.
- Brockhausen, I. and Stanley, P. (2015) 'O-GalNAc glycans', in Varki, A. et al. (eds.) *\*Essentials of Glycobiology\**. 3rd edn. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
- Chen, D., Zhang, X., Xu, S. and Liu, D. (2026) 'Current CMC challenges in oligonucleotide API development: from IND to NDA', *\*Medicinal Chemistry Research\**, 35, pp. 711–725.
- Collotta, D., Bertocchi, I., Chiapello, E. and Collino, M. (2023) 'Antisense oligonucleotides: a novel frontier in pharmacological strategy', *Frontiers in Pharmacology*, 14, article 1304342. Available at: <https://doi.org/10.3389/fphar.2023.1304342>.
- Crooke, S.T., Baker, B.F., Crooke, R.M. and Liang, X.H. (2021) 'Antisense technology: an overview and prospectus', *Nature Reviews Drug Discovery*, 20, pp. 427–453.
- European Medicines Agency (EMA) (2024) \*Guideline on the Development and Manufacture of Oligonucleotides\*. Draft guideline. EMA/CHMP/CVMP/QWP/262313/2024. Amsterdam: European Medicines Agency.
- Geary, R.S. et al. (2015) 'Clinical and preclinical pharmacokinetics and pharmacodynamics of mipomersen', *Clinical Pharmacokinetics*, 54, pp. 133–146.
- Geary, R.S., Watanabe, T.A., Truong, L., Freier, S., Lesnik, E.A., Sioufi, N.B., Sasmor, H., Manoharan, M. and Levin, A.A. (2001) 'Pharmacokinetic properties of 2'-O-(2-

- methoxyethyl)-modified oligonucleotide analogs in rats', *Journal of Pharmacology and Experimental Therapeutics*, 296(3), pp. 890–897. Useful supporting reference for the PS/protein-binding/tissue-distribution point:
- Ionis Pharmaceuticals (2013) KYNAMRO (mipomersen sodium) prescribing information. FDA.
- Ionis Pharmaceuticals (2019) TEGSEDI (inotersen) prescribing information. FDA.
- Ionis Pharmaceuticals (2026) The science behind our RNA-targeted medicines. Ionis Pharmaceuticals.
- KEGG (2026) KEGG DRUG: Tofersen. *Kyoto Encyclopedia of Genes and Genomes*.
- Lovrić, J., Yan, J., Li, X., Karlsborn, T., Bood, M., Dahlén, A., Hilgendorf, C. and Bhatt, D.K. (2025) 'In vitro structure–activity relationship stability study of antisense oligonucleotide therapeutics using biological matrices and nucleases', *Pharmacology Research & Perspectives*, 13(3). doi: 10.1002/prp2.70096.
- Migawa, M.T., Shen, W., Wan, W.B., Vasquez, G., Oestergaard, M.E., Low, A., De Hoyos, C.L., Gupta, R., Murray, S., Tanowitz, M., Bell, T.A., Nichols, J.G., Gaus, H., Swayze, E.E., Crooke, S.T. and Seth, P.P. (2021) 'Towards next generation antisense oligonucleotides: mesylphosphoramidate modification improves therapeutic index and duration of effect of gapmer antisense oligonucleotides', *Nucleic Acids Research*, 49(16), pp. 9026–9041. The study reports that PS/MsPA combinations improved potency/safety, reduced inflammatory effects and extended duration of effect.
- Nair, J.K. et al. (2014) 'Targeted delivery of antisense oligonucleotides to hepatocytes using N-acetylgalactosamine', *Nucleic Acids Research*, 42(13), pp. 8796–8807.
- Novartis (2021) \*FDA approves Novartis Leqvio® (inclisiran), first-in-class siRNA to lower cholesterol and keep it low with two doses a year\*. Basel: Novartis.
- Roberts, T.C., Langer, R. and Wood, M.J.A. (2020) 'Advances in oligonucleotide drug delivery', *Nature Reviews Drug Discovery*, 19, pp. 673–694.
- Rodríguez, E. et al. (2023) 'Molecular recognition of GalNAc in mucin-type O-glycosylation', *\*Accounts of Chemical Research\**, 56(6), pp. 744–756.
- Rodrigues, M. and Yokota, T. (2023) 'An overview of recent US-approved gene therapies for Duchenne muscular dystrophy and their respective clinical development programs', *Drugs & Therapy Perspectives*.
- Sarepta Therapeutics (2016) EXONDYS 51 (eteplirsen) prescribing information. FDA.
- Stetsenko, D.A. (2023) 'Mesyl phosphoramidate oligonucleotides: a new promising type of antisense agents', in *Handbook of Chemical Biology of Nucleic Acids*. Springer Nature. This chapter describes MsPA/ $\mu$  linkages as negatively charged phosphodiester mimics with high nuclease resistance and RNase H-compatible antisense potential.
- Stockert, R.J. (1995) 'The asialoglycoprotein receptor: relationships between structure, function, and expression', *\*Physiological Reviews\**, 75(3), pp. 591–609.
- Sutton, J.M., et al. (2025) 'In vitro structure–activity relationship stability study of antisense oligonucleotide therapeutics using biological matrices and nucleases', *Pharmacology Research & Perspectives*, 13(3), e70096. Available at: <https://DOI.ORG/10.1002/PRP2.70096>
- U.S. Food and Drug Administration (2019) \*FDA approves givosiran for acute hepatic porphyria\*. Silver Spring, MD: FDA.
- U.S. Food and Drug Administration (2020) \*Oxlumo prescribing information\*. Silver Spring, MD: FDA.
- U.S. Food and Drug Administration (2025) \*Leqvio prescribing information\*. Silver Spring, MD: FDA

- U.S. Food and Drug Administration (FDA) (2021) \*IND Submissions for Individualized Antisense Oligonucleotide Drug Products for Severely Debilitating or Life-Threatening Diseases: Chemistry, Manufacturing, and Controls Recommendations\*. Draft guidance for sponsor-investigators. Silver Spring, MD: Center for Drug Evaluation and Research.
- Witztum, J.L. et al. (2019) 'Volanesorsen and triglyceride levels in familial chylomicronaemia syndrome', *New England Journal of Medicine*, 381, pp. 531–542.

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